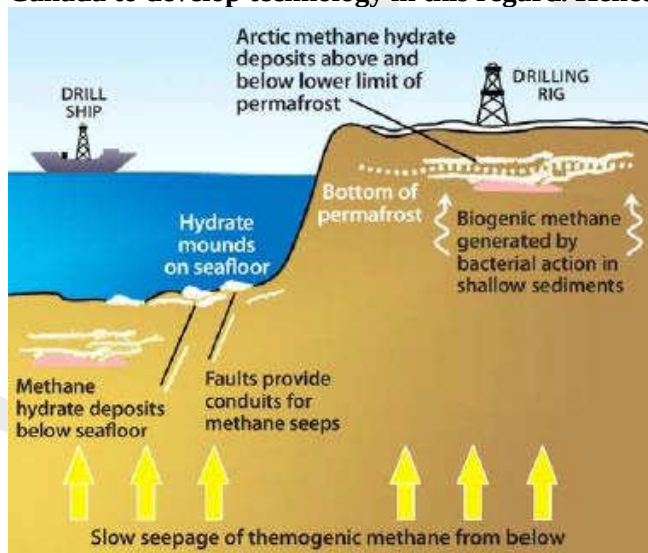


ANSWERS & EXPLANATIONS

GENERAL STUDIES (P) TEST – 3788 (2023)

Q 1.D

- **Methane Hydrate is a cage-like lattice of ice inside which are trapped molecules of methane, the chief constituent of natural gas. Several other names are commonly used for methane hydrate. These include methane clathrate, hydromethane, methane ice, fire ice, natural gas hydrate, and gas hydrate. Most methane hydrate deposits also contain small amounts of other hydrocarbon hydrates. Hence statement 1 is correct.**
- **It is found in a sea-bed that forms at low temperatures and high pressure. Hence statement 2 is correct.**
- Four Earth environments have the temperature and pressure conditions suitable for the formation and stability of methane hydrate. These are:
 - sediment and sedimentary rock units below Arctic permafrost;
 - sedimentary deposits along continental margins;
 - deep-water sediments of inland lakes and seas; and,
 - under Antarctic ice.
- **It is found in onshore deposits in the permafrost of northern Canada and Russia. Heating the deposits or lowering the pressure will release gas from the solid. One liter of solid hydrate releases around 165 liters of gas. Apart from the US and Japan, India has entered into an agreement with Canada to develop technology in this regard. Hence statement 3 is correct.**



Q 2.B

- **Major Industrial Regions (There are 8 such regions)**
 - Mumbai-Pune Region,
 - Hugli Region,
 - Bangalore-Tamil Nadu Region,
 - Gujarat Region,
 - Chota Nagpur Region,
 - Vishakhapatnam-Guntur Region,
 - Gurugram-Delhi-Meerut Region, and
 - Kollam-Thiruvantapuram Region.

- **Minor Industrial Regions (There are 13 such regions)**

- Ambala-Amritsar,
- Saharanpur-Muzaffarnagar-Bijnor,
- Indore-Dewas-Ujjain,
- Jaipur-Ajmer,
- Kolhapur-South Kannada,
- **Northern Malabar,**
- Middle Malabar,
- Adilabad-Nizamabad,
- **Prayagraj-Varanasi-Mirzapur,**
- Bhojpur-Munger,
- Durg-Raipur,
- Bilaspur-Korba, and
- Brahmaputra valley

Q 3.B

- Economic activities result in the production of goods and services while sectors are the group of economic activities classified on the basis of some criteria. **They are broadly grouped into primary, secondary, tertiary and quaternary activities.**
- Primary activities are directly dependent on environment as these refer to utilisation of earth's resources such as land, water, vegetation, building materials and minerals. It, thus includes, hunting and gathering, pastoral activities, fishing, forestry, agriculture, and mining and quarrying. **People engaged in primary activities are called redcollar workers due to the outdoor nature of their work.**
- Secondary activities add value to natural resources by transforming raw materials into valuable products. **Secondary activities, therefore, are concerned with manufacturing, processing and construction (infrastructure) industries. People engaged in secondary activities are called blue collar workers. Hence option (b) is the correct answer.**
- **Quinary Activities is the part of the economy where the top-level decisions are made.** It focus on the creation, re-arrangement and interpretation of new and existing ideas; data interpretation and the use and evaluation of new technologies. **Profession under this category often referred as 'gold collar' professions,** they represent another subdivision of the tertiary sector representing special and highly paid skills of senior business executives, government officials, research scientists, financial and legal consultants, etc.
- Quaternary activities involve some of the following: the collection, production and dissemination of information or even the production of information. **Quaternary activities centre around research, development and may be seen as an advanced form of services involving specialised knowledge and technical skills.**

Q 4.B

- **The major tribes of equatorial regions are Indian tribes of the Amazon basin, Pygmies of the Congo Basin, and Orang Asli of Malaysia. The Bedouins are nomadic Arab tribes who have historically inhabited the desert regions in the Arabian Peninsula. Hence statement 1 is not correct.**
- The equatorial regions are generally sparsely populated. In the forests, most primitive people live as **hunters and collectors and the more advanced ones practice shifting cultivation. Hence statement 2 is correct.**

Q 5.C

- Coal is a combustible black or brownish-black sedimentary rock with a high amount of carbon and hydrocarbons. Coal is classified as a nonrenewable energy source because it takes millions of years to form.
- **Types of coal are:**
 - **Lignite** - often referred to as brown coal, is a soft brown combustible sedimentary rock that is formed from naturally compressed peat. **It is considered the lowest rank of coal due to its relatively low heat content. It has a carbon content of around 25-35%, a high inherent moisture content sometimes as high as 66%, and an ash content ranging from 6% to 19%. It is mined in Bulgaria, Kosovo, Greece, Germany, Poland, Serbia, Russia, Turkey, the United States, Canada, India, Australia, and many other parts of Europe, and it is used almost exclusively as a fuel for steam-electric power generation. Hence statement 1 is correct.**
 - **Bituminous coal or black coal is relatively soft coal containing a tar-like substance called bitumen. It is of higher quality than lignite coal but of poorer quality than anthracite. The**

carbon content of bituminous coal is around 60-80%; the rest is composed of water, air, hydrogen, and sulfur. **Hence statement 2 is not correct.**

- **Anthracite is a hard, compact variety of mineral coal that has a high luster. It has the highest carbon content, the fewest impurities, and the highest calorific content of all types of coal. The carbon content is between 92.1% and 98%.** It is used mainly in power generation, in the metallurgy sector. Anthracite accounts for about 1% of global coal reserves,[4] and is mined in only a few countries around the world. China accounts for the majority of global production; other producers are Russia, Ukraine, North Korea, Vietnam, the UK, Australia, and the US. **Hence statement 3 is not correct.**

Q 6.C

- Wheat is the second most important cereal crop in India after rice. It is primarily a crop in the temperate zone. Hence, its cultivation in India is done during winter i.e. rabi season.
- This rabi crop requires a cool growing season and bright sunshine at the time of ripening. There are two important wheat-growing zones in the country – the Ganga-Satluj plains in the northwest and the black soil region of the Deccan. The major wheat-producing states are Punjab, Haryana, Uttar Pradesh, Madhya Pradesh, Bihar, and Rajasthan.
- **Wheat needs a winter temperature of 10° to 15° C and a summer temperature of 21°C to 26°C during ripening to produce a good crop. Hence statement 3 is correct.**
- **It requires a rainfall of 50 to 75 cm.** Excessive rainfall is harmful to the wheat crop. The roots of the plant are destroyed in standing water. **Hence statement 1 is not correct.**
- **Light loamy soil is ideal for wheat cultivation. Hence statement 2 is correct.**
- Hence, its cultivation in India is done during winter i.e. rabi season. **Cultivation of wheat is not labor-intensive.**
- About 85 percent of the total area under this crop is concentrated in the north and central regions of the country i.e. Indo-Gangetic Plain, Malwa Plateau, and Himalayas up to 2,700 m altitudes. Being a rabi crop, it is mostly grown under irrigated conditions. **But it is a rainfed crop in Himalayan highlands and parts of the Malwa plateau in Madhya Pradesh. About 14 percent of the total cropped area in the country is under wheat cultivation.**
- Uttar Pradesh, Madhya Pradesh, Punjab, Haryana, and Rajasthan are leading wheat-producing states. The yield level of wheat is very high (above 4,000 k.g. per ha) in Punjab and Haryana whereas, Uttar Pradesh, Rajasthan, and Bihar have moderate yields. The states like Madhya Pradesh, Himachal Pradesh, and Jammu, and Kashmir growing wheat under rainfed conditions have low yields.

Q 7.D

- A port is a maritime facility that may comprise one or more wharves where ships may dock to load and discharge passengers and cargo. Ports are extremely important to the global economy, responsible for 70% of global merchandise trade by value.
- **Important Ports of the World:**
 - Shanghai, China
 - **Havana - Cuba**
 - Qingdao, China
 - Jebel Ali, Dubai, United Arab Emirates
 - **Rotterdam, The Netherlands**
 - Port Klang, Malaysia
 - **Antwerp - Belgium**
 - Kaohsiung, Taiwan, China
 - Los Angeles, U.S.A
 - **Hamburg, Germany**
 - **Baltimore, U.S.A.**
 - Keihin Ports, Japan
 - New York-New Jersey, U.S.A.
 - Colombo, Sri Lanka
 - Hambantota, Sri Lanka
 - Ho Chi Minh City, Vietnam
 - Hanshin Port, Japan
 - Jawaharlal Nehru Port (Nhava Sheva), India
 - Piraeus, Greece
 - Algeciras, Spain

- Jeddah, Saudi Arabia
- **Santos, Brazil**
- Fuzhou, China
- Vancouver, Canada
- Salalah, Oman
- Khasab Port, Oman
- Chabahar, Iran
- Port of Bassein, Myanmar
- Port of Karachi, Pakistan
- Gwadar Port, Pakistan
- Hodeidah Port, Yemen
- Aden Port, Yemen
- **Hence only pairs 2 and 4 are not correctly matched.**

Q 8.B

- Nomadic herding or pastoral nomadism is a type of subsistence in which herders primarily rely on domesticated animals for food and livelihood. They generally move from one place to another in search of pasture for their animals. There are three core regions of pastoral nomadism. These regions are the Atlantic shores of North Africa eastwards across the Arabian Peninsula into Mongolia and central China. The second region extends over the Tundra region of Eurasia. The third region is a small part of the southern hemisphere in South West Africa and on the island of Madagascar.
- **Llamas are reared as part of pastoral nomadism in the Andes region. Prairies lie in North America. Hence statement 1 is not correct.**
- **Reindeer are reared in the Arctic and sub-Arctic regions as part of pastoral nomadism. Hence statement 2 is correct.**

Q 9.A

- **Recent context: Scientists from Agharkar Research Institute (ARI), an autonomous institute of the Department of Science and Technology have developed and patented a diagnostic tool that detects the White Spot Syndrome Virus (WSSV).**
- **White spot disease (WSD) is a highly contagious viral infection of crustaceans that can cause high levels of mortality in cultured shrimp.** Since its first outbreak in 1992–93, this disease has caused serious economic losses. The causative agent of WSD is the white spot syndrome virus (WSSV), which is a large, nonoccluded, enveloped, rod-shaped elliptical DNA virus with a tail-like extension at one end. **Hence statement 1 is correct.**
- **The virus infects only crustaceans** (prawns, lobsters, crabs, etc) and appears not to be related to any other known viruses. **Hence statement 2 is correct.**
- **At present, there is no treatment available** to interfere with the unrestrained occurrence and spread of the disease. **Hence statement 3 is correct.**

Q 10.C

- **Sub Urbanisation is a new trend of people moving away from congested urban areas to cleaner areas outside the city in search of a better quality of living.** Important suburbs develop around major cities and everyday thousands of people commute from their homes in the suburbs to their work places in the city. Like in case of Delhi, people residing in Gautam Buddha Nagar (Noida) and commuting everyday to work in their offices based in Delhi.
- **Megalopolis:** This Greek word meaning “great city”, was popularised by Jean Gottman (1957) and signifies ‘super- metropolitan’ region extending, as union of conurbations. The urban landscape stretching from Boston in the north to south of Washington in U.S.A. is the best known example of a megalopolis.
- **Hence, option (c) is the correct answer.**

Q 11.B

- The coal deposits although exist in nearly every part of the world, but commercially exploitable coal reserves are found mainly in China, USA, India, Australia, Indonesia, Russia, Canada, South Africa, Columbia, Kazakhstan, and Ukraine.
- **Major coalfields of the world are:**
 - **North America** - Pennsylvania anthracite field, Appalachian bituminous field, Eastern Illinois field – Illinois, Indiana and Kentucky, Western interior field – Iowa, Missouri, Oklahoma, Gulf province – Texas, Alabama and Arkansas, Rocky mountain province- Utah, Colorado, Wyoming, Montana, new Mexico, Canada – Prairies, British Columbia coalfields, Nova Scotia Coal fields.

- **Europe** - Donetsk coal basin (anthracite and high-grade bituminous coal), Moscow-Tula coalfields, Kuznetsk coal basin, Karaganda field, Silesia coal fields, **Ruhr area of Germany**, Other coal fields in Urals, Taimyr fields of the Arctic, deposits of the Caucasus mountains.
- **Asia** - China – Shanxi, Fushun, Inner Mongolia, Kansu Japan – Chikugo coalfield, Ishikari coalfield, **India – Damodar valley, Raniganj, Bokaro, Jharia, Singareni**. Pakistan - Quetta, Kalabagh and Thar coalfields Australia – Bowen Basin coalfield, Galilee Basin coalfield, South Maitland coalfield, Sydney Basin coalfield, and Latrobe valley coalfield.
- **Africa** - Transvaal and Natal – Middleburg, Vereeniging and Witbank, **Zimbabwe – Wankie**, Zaire – Luena, Mozambique – Maniamba, Zambia – Nkandabwe and Mamba, Nigeria – Enugu.
- **South America** - **Brazil – Santa Catarina and Rio grande de sul**, Chile – Concepcion, Columbia – Cauca valley coalfield, Mexico – Piedras Negras, Sabinas and Lampazos.
- **Hence option (b) is the correct answer.**

Q 12.C

- **This is the population pyramid of Japan**, it has a narrow base and tapered top. It indicates low birth and death rate. The population growth in developed countries like Japan is almost zero or negative. **Hence option (c) is the correct answer.**

Q 13.A

Age-Sex Pyramid:

The age-sex structure of a population refers to the number of females and males in different age groups. A population pyramid is used to show the age-sex structure of the population. The shape of the population pyramid reflects the characteristics of the population. The left side shows the percentage of males while the right side shows the percentage of women in each age group.

Expanding Populations:

The age-sex pyramid of Nigeria for example is a **triangular shaped** pyramid with a wide base and is typical of less developed countries. These have larger populations in lower age groups due to high birth rates.

Ex: Bangladesh and Mexico etc.

Constant Population:

In the case of Australia age-sex pyramid is **bell shaped** and tapered towards the top. This shows birth and death rates are almost equal leading to a near constant population.

Declining Populations:

The Japan pyramid has a narrow base and a tapered top showing low birth and death rates. The population growth in developed countries is usually zero or negative.

Q 14.D

- **Recent Context: EXIM Bank plans to raise around \$3 billion in 2022-23 (FY23) via overseas bonds to support fresh lending and refinance a portion of the old debt.**
- **About EXIM Bank:**
 - Established by the Government of India, under the **Export-Import Bank of India Act, 1981**. as a purveyor of export credit, mirroring global Export Credit Agencies. Exim Bank serves as a growth engine for industries and SMEs through a wide range of products and services. This includes import of technology and export product development, export production, export marketing, pre-shipment and post-shipment, and overseas investment. **Hence statement 1 is not correct.**
 - **It extends Lines of Credit (LOCs) to overseas financial institutions, regional development banks, sovereign governments, and other entities overseas. Hence statement 2 is not correct.**
 - The objective of the bank is to enable buyers in those countries to import developmental and infrastructure projects, goods, and services from India, on deferred credit terms.

Q 15.A

- **Push factors are negative things that make people want to move to a new area.** Reasons include less economic opportunities, poor living conditions, political turmoil, unpleasant climate, epidemics, natural disasters, etc. **Hence options 1 and 2 are correct.**

- Pull factors are positive aspects that attract people to move to a place. It includes better job opportunities and living conditions, peace and stability, security of life and property, and a pleasant climate. **Hence options 3 and 4 are not correct.**

Q 16.C

- **India is poorly endowed with non-ferrous metallic minerals except bauxite.**
- Bauxite is the ore, which is used in the manufacturing of aluminum.
- **Bauxite is found mainly in tertiary deposits and is associated with laterite rocks occurring extensively either on the plateau or hill ranges of peninsular India and also in the coastal tracts of the country.**
- **It is not specifically a mineral but a rock consisting mainly of hydrated aluminum oxides.**
- India is self-sufficient in bauxite reserves.
- **Odisha happens to be the largest producer of Bauxite. Kalahandi and Sambalpur are the leading producers.** The other two areas which have been increasing their production are Bolangir and Koraput.
- **Hence option (c) is the correct answer.**

Metal	Name of ore
sodium	rock salt
calcium	limestone
magnesium	magnesite
aluminium	bauxite
zinc	zinc blende
iron	haematite magnetite
tin	cassiterite
lead	galena
copper	copper pyrite or chalcopyrite

Deposits and Producers of Bauxite in India:

- The peatlands of Lohardaga in Jharkhand have rich deposits.
- Gujarat, Chhattisgarh, Madhya Pradesh, and Maharashtra are other major producers.
- Tamil Nadu, Karnataka, and Goa are minor producers of bauxite.
- **Odisha alone accounts for 51% of the country's resources of bauxite followed by Andhra Pradesh (16%), Gujarat (9%), Jharkhand (6%), Maharashtra (5%), and Madhya Pradesh & Chhattisgarh (4% each).** Major bauxite resources are concentrated in the East Coast bauxite deposits in Odisha and Andhra Pradesh
- The production of bauxite at 21.82 million tonnes in 2019-20 decreased by 8% as compared to the previous year. During the year under review, Odisha accounted for 71% of the total output followed by Gujarat (9%) and Jharkhand (6%).

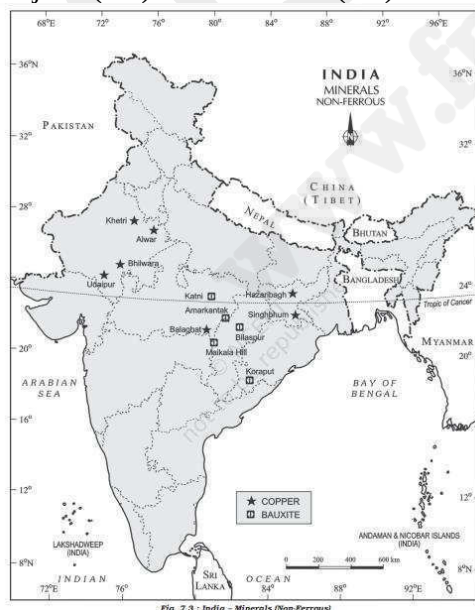


Fig. 7.3 : India - Minerals (Non-Ferrous)

Bauxite reserves in the world:

- Australia continued to be the major producer and accounted for about 29% share of the total production, followed by China (19%), Guinea (18%), Brazil (10%), and India (7%).

Q 17.D

- One of the important achievements of the Indian Railways has been the construction of the Konkan Railway in 1998. **It is a 760 km long rail route connecting Roha in Maharashtra to Mangalore in Karnataka.**
- It is considered an engineering marvel. It crosses 146 rivers, streams, nearly 2000 bridges, and 91 tunnels.
- **The 756.25 km (469.91 mi) long railway line connects the states of Maharashtra, Goa, and Karnataka. Tamil Nadu is not covered under the Konkan Railways network.**
- Indian Railways (IR) is one of the world's largest rail networks with 67,956 Route Kilometres of route lengths as of 31.03.2020. **Out of 67,956 RKM, Broad Gauge constitutes 63,949 RKM (94.10%), Metre Gauge 2,402 RKM (3.54%) and Narrow Gauge 1,809 RKM (2.36%).**

Q 18.A

- Ports of Call: These are the ports which originally developed as calling points on main sea routes where ships were used to anchor for refueling, watering, and taking food items. Later on, they developed into commercial ports. Aden, Honolulu, and Singapore are good examples.
- Packet Station: These are also known as **ferry ports. These packet stations are exclusively concerned with the transportation of passengers and mail across water bodies covering short distances.** These stations occur in pairs located in such a way that they face each other across the water body, e.g. Dover in England and Calais in France across the English Channel. **Hence statement 1 is correct.**
- **Naval Ports: These are ports that have only strategic importance.** These ports serve warships and have repair workshops for them. Kochi and Karwar are examples of such ports in India. **Hence statement 2 is correct.**
- **Entrepot Ports: These are collection centers where the goods are brought from different countries for export.** Singapore is an entrepot for Asia. Rotterdam for Europe, and Copenhagen for the Baltic region. **Hence statement 3 is not correct.**

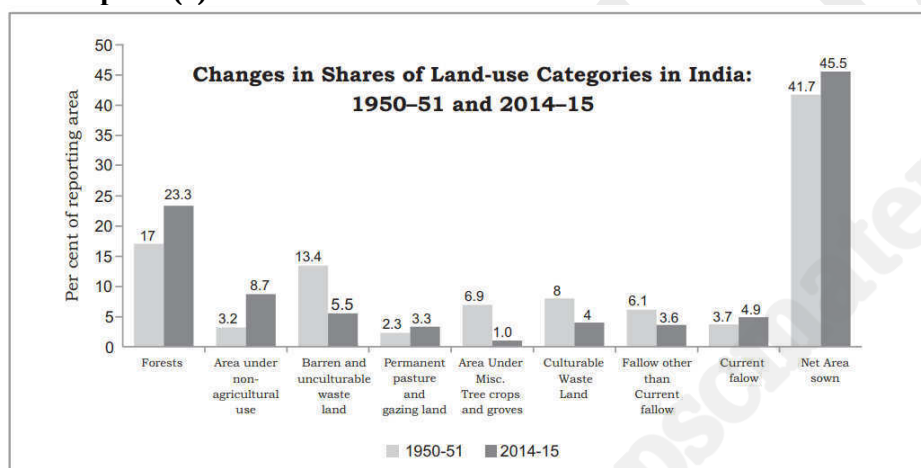
Q 19.C

- **Recent Context: Centre opposes Odisha's Ordinance for Landmark Lingaraj Temple.**
- **The Lingaraj Temple Ordinance of 2020** was introduced by the Odisha government to manage the rituals and other activities of the temple and eight other associated temples.
- The Ministry of Home Affairs contended that since the ordinance covers 12 centrally protected monuments including the Lingaraj temple and three tanks, it was outside the legislative competence of the state legislature as it violates the provisions of Ancient Monuments and Archaeological Sites and Remains (AMASR) Act, 1958.
 - **The AMASR Act** provides for the preservation of ancient and historical monuments and archaeological sites and remains of national importance.
- The ordinance also provides for the repair and construction of new buildings while the center contended that construction can only be allowed by the National Monuments Authority.
- **Lingaraj Temple:**
 - It is dedicated to Lord Shiva and is one of the oldest in Bhubaneswar, was constructed by King Jajati Keshari in the 10th Century and completed by King Lalatendu Keshari in the 11th Century. **Hence statement 1 is correct.**
 - Built-in red stone, it is a classic example of the Kalinga style of architecture. This style broadly comes under the Nagara style. **Hence statement 2 is correct.**
 - It is built in the Deula style that has four components, namely-
 - Garbha Griha (sanctum sanctorum),
 - Yajna Shala (hall for prayers),
 - Bhoga Mandap (hall of offering) and
 - Natya Shala (hall of dance).

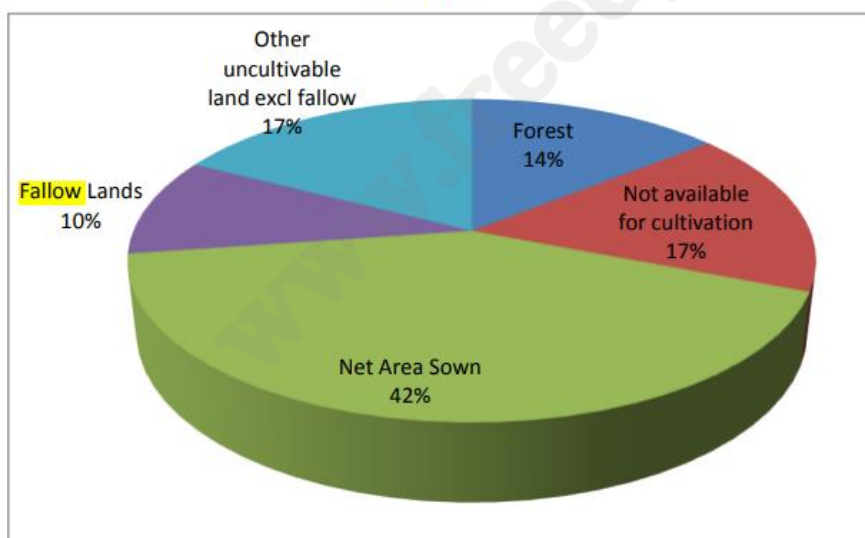
Q 20.A

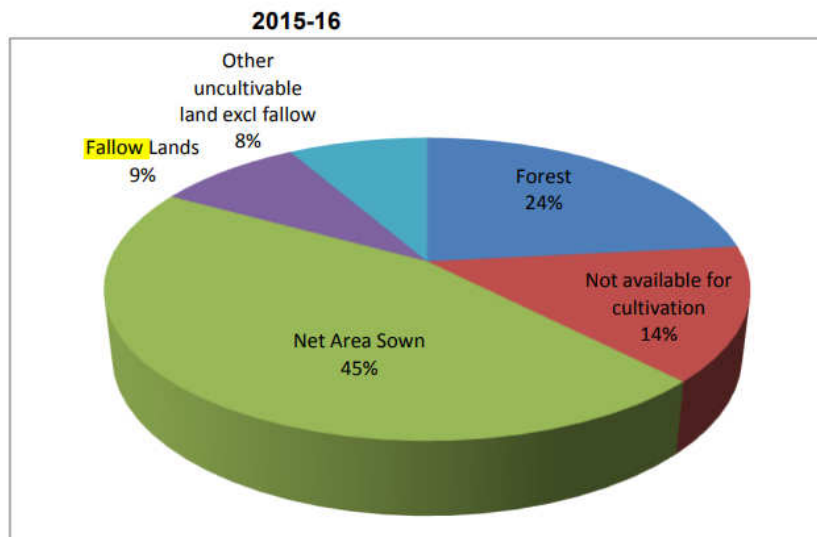
- India has undergone major changes within the economy over the past four or five decades, and this has influenced the **land-use changes in the country.**
- Four categories have undergone increase, while four have registered decline. Share of the area under forest, the area under non-agricultural uses, current fallow lands, and the net area sown have shown an increase.

- **The following observations can be made about these increases:**
 - The rate of increase is the highest in the case of areas under non-agricultural uses. This is due to the changing structure of the Indian economy, which is increasingly dependent on the contribution of industrial and services sectors and the expansion of related infrastructural facilities. Also, an expansion of area under both urban and rural settlements has added to the increase. **Thus, the area under non-agricultural uses is increasing at the expense of wastelands and agricultural land.**
 - The increase in the share under forest, as explained before, can be accounted for by an increase in the demarcated area under forest rather than an actual increase in the forest cover in the country.
 - **The increase in the current fallow cannot be explained from information pertaining to only two points.** The trend of current fallow fluctuates a great deal over years, depending on the variability of rainfall and cropping cycles.
 - The increase in net area sown is a recent phenomenon due to the use of culturable wasteland for agricultural purposes. Before which it was registering a slow decrease. There are indications that most of the decline had occurred due to the increases in area under nonagricultural use. (Note: the expansion of building activity on agricultural land in your village and city).
 - **The four categories that have registered a decline are barren and wasteland, culturable wasteland, the area under pastures and tree crops, and fallow lands.**
- **The following explanations can be given for the declining trends:**
 - As the pressure on land increased, both from the agricultural and nonagricultural sectors, the **wastelands and culturable wastelands have witnessed a decline over time.**
 - The decline in **land under pastures and grazing lands can be explained by pressure from agricultural land.** Illegal encroachment due to expansion of cultivation on common pasture lands is largely responsible for this decline.
- **Hence option (a) is the correct answer.**



1950-51





Q 21.C

- **Agriculture is a purely land-based activity, unlike secondary and tertiary activities.** In other words, the contribution of land to agricultural output is more compared to its contribution to the outputs in the other sectors. Thus, lack of access to land is directly correlated with the incidence of poverty in rural areas
- Quality of land has a direct bearing on the productivity of agriculture, which is not true for other activities.
- **As the scope for bringing in additional land under the net sown area in India is limited. There is, thus, an urgent need to evolve and adapt land-saving technologies.**
 - Such technologies can be classified under two heads – those which raise the yield of any particular crop per unit area of land and those which increase the total output per unit area of land from all crops grown over one agricultural year by increasing land-use intensity.
 - **The advantage of the latter kind of technology is that along with increased output from limited land, it also increases the demand for labor significantly. Hence statement 2 is correct**
- For a land-scarce but labor-abundant country like India, a high cropping intensity is desirable not only for fuller utilization of land resources but also for reducing unemployment in the rural economy.
- **Cropping Intensity is the ratio of Gross Area Sown to the Net Area Sown. Hence statement 1 is correct.**
 - Around 51% of India's geographical area is already under cultivation as compared to 11% of the world average. The present cropping intensity of 136% has registered an increase of only 25% since independence.

Q 22.D

Intensive Subsistence Agriculture

- This type of agriculture is largely found in densely populated regions of monsoon Asia.
- This type of farming is practised in areas of **high population pressure on land**. Though the 'right of inheritance' leading to the division of land among successive generations has rendered land-holding size uneconomical, the farmers continue to take the maximum output from the limited land in the absence of an alternative source of livelihood. Thus, there is enormous pressure on agricultural land. **Hence option 1 is correct.**
- It is labour intensive farming, where high doses of biochemical inputs and irrigation are used for obtaining higher production. **Hence option 2 is correct.**

There are two types of intensive subsistence agriculture.

- Intensive subsistence agriculture dominated by wet paddy cultivation:
 - This type of agriculture is characterised by the **dominance of the rice crop**. Landholdings are very small due to the high density of the population. Farmers work with the help of family labour leading to intensive use of land. The use of **machinery is limited and most of the agricultural operations are done by manual labour**. **Hence option 3 is not correct.**
 - Farmyard manure is used to maintain the fertility of the soil. In this type of agriculture, **the yield per unit area is high but per labour productivity is low**. **Hence statement 4 is correct.**
- Intensive subsistence agriculture dominated by crops other than paddy:
 - Due to the difference in relief, climate, soil and some other geographical factors, it is not practical to grow paddy in many parts of monsoon Asia.
 - Wheat, soyabean, barley and sorghum are grown in northern China, Manchuria, North Korea and North Japan. **In India wheat is grown in western parts of the Indo-Gangetic plains and millets**

are grown in dry parts of western and southern India. Most of the characteristics of this type of agriculture are similar to those dominated by wet paddy except that irrigation is often used.

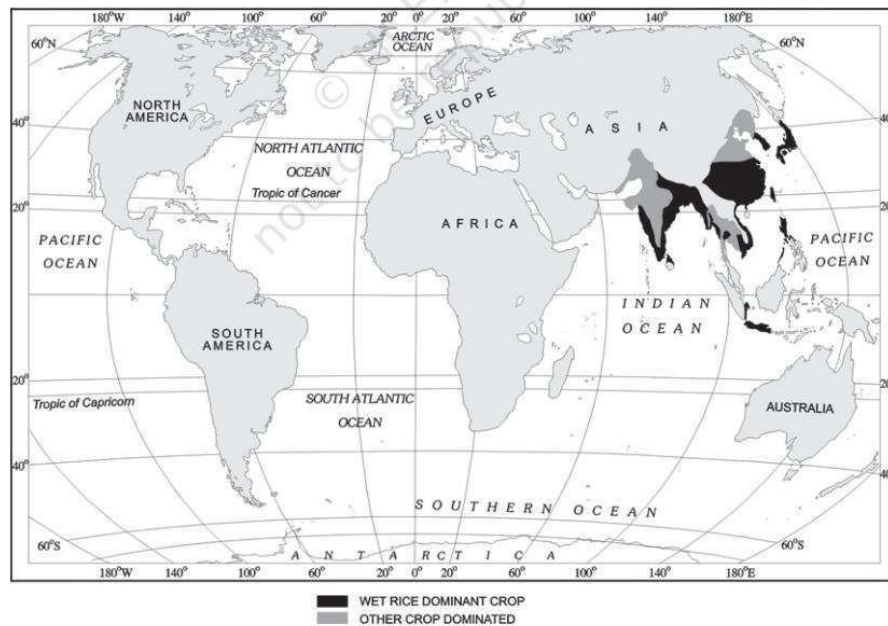


Fig. 5.8: Areas of Intensive Subsistence Farming

Q 23.A

- **Weight losing raw materials are materials that weigh less after getting manufactured than what should have been their weight as a raw material.** For example, copper was taken from its raw material copper ore.
- Industries using weight-losing raw materials are located in the regions where raw materials are located. The **sugar mills** in India are located in sugarcane growing areas.
- Similarly, the locations of the **pulp industry, copper smelting, and pig iron industries are located near their raw materials.**
- In the **iron and steel industries, iron ore and coal both are weight-losing raw materials. Therefore Iron and Steel Industry, Pulp Industry, and Sugar mills Industry are categorized as a weight-losing raw materials-based industry.**
- Therefore, an optimum location for iron and steel industries should be near raw material sources. This is why most of the iron and steel industries are located either near coalfields (Bokaro, Durgapur, etc.) or near sources of iron ore (Bhadrahati, Bhilai, and Rourkela).
- Similarly, industries based on perishable raw materials are also located close to raw material sources.
- Markets provide outlets for manufactured products. Heavy machines, machine tools, and heavy chemicals are located near the high-demand areas as these are market-orientated.
- **The cotton textile industry uses a non-weight-losing raw material** and is generally located in large urban centers, e.g. Mumbai, Ahmedabad, Surat, etc.
- **Petroleum refineries are also located near the markets** as the transport of crude oil is easier and several products derived from them are used as raw materials in other industries. **This industry is also a non-weight-losing raw material-based industry.** Koyali, Mathura, and Barauni refineries are typical examples. Ports also play a crucial role in the location of oil refineries.
- **Hence option (a) is the correct answer.**

Q 24.C

- Demographic transition theory can be used to describe and predict the future population of any area. **Hence statement 1 is correct.** The theory tells us that the population of any region changes from high births and high deaths to low births and low deaths as society progresses from rural agrarian and illiterate to urban industrial and literate society. **Hence statement 2 is correct.** These changes occur in the stages which are collectively known as the demographic cycle.
- Three Stage demographic transition theory:

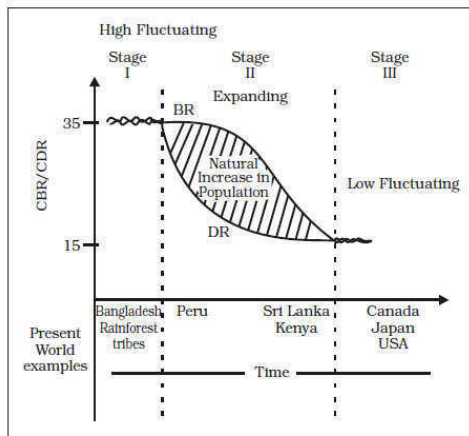


Fig. 2.5: Demographic Transition Theory

- The first stage has high fertility and high mortality because people produce more to compensate for the deaths due to epidemics and variable food supply. The population growth is slow and most of the people are engaged in agriculture where large families are an asset. Life expectancy is low, people are mostly illiterate and have low levels of technology.
- The fertility rate remains high at the beginning of the second stage but it declines with time. This is accompanied by a reduced mortality rate. Improvements in sanitation and health conditions lead to a decline in mortality. Because of this gap, the net addition to the population is high.
- In the last stage, both fertility and mortality decline considerably. The population is either stable or grows slowly. The population becomes urbanized, literate and has the high technical know-how, and deliberately controls the family size. **Hence statement 3 is not correct.**

Q 25.A

- **Recently, Cabinet Committee on Economic Affairs approved the distribution of fortified rice across Government Schemes by 2024 in a phased manner with an aim to combat nutrition deficiency in all states and UTs.**
 - Supply of fortified rice through the Targeted Public Distribution System (TPDS) will be done under National Food Security Act (NFSA), Integrated Child Development Services (ICDS), PM POSHAN, and other welfare schemes (OWS).
 - **The entire cost of rice fortification would be borne by the Centre. Hence statement 1 is correct.**
- Three phases are envisaged for full implementation:
 - Phase-I: Covering ICDS and PM POSHAN in India all over by March 2022 which is under implementation.
 - Phase-II: Phase I above plus TPDS and OWS in all Aspirational and High Burden Districts on stunting (total 291 districts) by March 2023.
 - Phase-III: Phase II above plus covering the remaining districts of the country by March 2024.
- **Rice fortification is a process of adding micronutrients, commonly iron, Vitamin B12, and folic acid, to regular rice to improve the dietary requirement.**
 - **In India, rice is fortified using extrusion technology. Hence statement 2 is not correct.**
 - In this technology, milled rice is pulverized and mixed with a premix containing vitamins and minerals.
 - Fortified rice kernels (FRK) are produced from this mixture using an extruder machine.
 - **FRK is added to traditional rice in ratios ranging from 1:50 to 1: 200 resulting in fortified rice nearly identical to traditional rice in aroma, taste, and texture.**
- **Benefits of rice fortification:**
 - Provide a nutritional boost to undernourished and vulnerable populations.
 - Combat anemia.
 - Cost-effective.
 - Helpful in pregnancy for the growth of the baby.

Q 26.C

- On the basis of size, capital investment and labour force employed, industries are classified as large, medium, small scale, and cottage industries.
- **On the basis of the system of large scale manufacturing, the world's major industrial regions may be grouped under two broad types, namely (i) traditional large-scale industrial regions which are**

thickly clustered in a few more developed countries. (ii) high-technology large scale industrial regions which have diffused to less developed countries.

- **Traditional Large-Scale Industrial Regions-** These are based on heavy industry, often located near coal-fields and engaged in metal smelting, heavy engineering, chemical manufacture or textile production. These industries are now known as smokestack industries. Hence option (c) is the correct answer.
- **Foot loose industries** can be located in a wide variety of places. **They are not dependent on any specific raw material, weight losing or otherwise.** They largely depend on component parts which can be obtained anywhere. They produce in small quantity and also employ a small labour force.
- **High technology**, or simply high-tech, is the latest generation of manufacturing activities. It is best understood as the application of intensive research and development (R and D) efforts leading to the manufacture of products of an advanced scientific and engineering character. Robotics on the assembly line, computer -aided design (CAD) and manufacturing, electronic controls of smelting and refining processes, and the constant development of new chemical and pharmaceutical products are notable examples of a high-tech industry. **High-tech industries which are regionally concentrated, self-sustained and highly specialised are called technopolies.**
- **Petrochemical industries are one of the fastest growing industries in India.** A variety of products come under this category of industries. Many items are derived from crude petroleum, which provide raw materials for many new industries; these are collectively known as petrochemical industries. **It is responsible for the manufacture and distribution of the various petrochemicals like polymers, chemicals, fibres and fibre intermediates.**

Q 27.C

- Rice is the staple crop of India, and every day millions of Indians find comfort in it. **With a high carbohydrate content, it is known to provide instant energy and is a staple that is consumed by the majority of India's population.**
 - It is a Kharif crop that requires high temperature, (above 25°C) and high humidity with annual rainfall above 100 cm. **Hence statement 2 is not correct.**
 - **In areas with less rainfall, it grows with the help of irrigation.** Rice is grown in the plains of north and north-eastern India, coastal areas, and the deltaic regions. The development of a dense network of canal irrigation and tubewells have made it possible to grow rice in areas of less rainfall such as Punjab, Haryana, western Uttar Pradesh, and parts of Rajasthan.
 - In southern states and West Bengal, the climatic conditions allow the cultivation of two or three crops of rice in an agricultural year. **In West Bengal farmers grow three crops of rice called 'aus', 'aman', and 'boro'. But in the Himalayas and northwestern parts of the country, it is grown as a Kharif crop during the Southwest Monsoon season.**
- **India contributes 21.9 percent of rice production in the world and ranked second after China in 2017.** About one-fourth of the total cropped area in the country is under rice cultivation. **Hence statement 3 is correct.**
- Rice being a tropical and sub-tropical plant requires a fairly high temperature of more than 22°C and an amount of rainfall of more than 100 cm. Irrigation is necessary for areas of lesser rainfall. Clayey alluvial soil in which water can remain standing is ideal for rice. **Hence statement 1 is not correct.**
- In India, rice is grown in almost half the states, with **West Bengal leading the way in terms of production with 14.71 million tonnes, followed by Uttar Pradesh (12.22 million tonnes) and Andhra Pradesh (11.57 million tonnes) as per the Agricultural Statistics 2014-15, Ministry of Agriculture & Farmers Welfare of the Government of India.**

Q 28.C

- Recently, Twitter has unanimously adopted a so-called shareholder rights plan, also known as a "poison pill". The Rights Plan will reduce the likelihood that any entity, person, or group gains control of Twitter through open market accumulation without paying all shareholders an appropriate control premium.
- **Poison pills give existing shareholders the option to buy more shares at a lower price, effectively diluting a new, hostile party's ownership stake.**
- **The poison pill strategy was developed by New York-based law firm Wachtell, Lipton, Rosen, and Katz in the 1980s.**
- **Types of poison pills:**
 - The flip-in and flip-over poison pill tactics are the two sorts of strategies. The flip-in option is the more popular of the two options.

- **Flip-in poison pill:** Allowing all shareholders, excluding the acquirer, to purchase additional shares at a discount is known as a flip-in poison pill strategy. While purchasing additional shares offers stockholders immediate profits, the strategy dilutes the value of the acquiring company's limited number of shares already owned.
- **Flip-over poison pill:** The flip-over strategy allows shareholders of the target company to purchase shares of the acquiring company at a significantly reduced price. Because it threatens to dilute and devalue the stock of the corporation attempting to take over the target, the tactic functions as a hostile takeover defense.
- **Hence option (c) is the correct answer.**

Q 29.A

- **Trans-Siberian Railway** was constructed for connecting European Russia to Siberia or Asian Russia and runs from St. Petersburg in the west to Vladivostok on the Pacific Coast in the east passing through Moscow, Ufa, Novosibirsk, Irkutsk, Chita, and Khabarovsk. It crosses eight time zones, being the longest railway in the world. Hence statement 1 is correct and statement 2 is not correct.
- **Trans-Manchurian Railways:** At 8,986 km (5,623 miles) from Moscow to Beijing, the Trans-Manchurian route is the longest that ends in China. Moscow to Beijing is the third-longest railway line, and it is 8986km long. Parts of the route go through the Trans-Siberian line, but it is still a different rail line.
- It is the most important route in Asia and the longest (9,332 km) double-tracked and electrified transcontinental railway in the world. **It has helped in opening up its Asian region to West European markets. Its development is on account of economic, political, and defense reasons.**

TRANS-SIBERIAN-RAILWAY



Q 30.B

- World crude steel production for the 64 countries reporting to the World Steel Association (world steel) was 174.4 million tonnes (Mt) in May 2021, a 16.5% increase compared to May 2020. Africa produced 1.4 Mt in October 2021, up 24.1% in October 2020.
- **The iron and steel industry has developed in Algeria, Egypt, Zimbabwe, and South Africa. South Africa is one of the major steel-producing countries in Africa. Egypt is the largest steel-producing country in Africa. The industry at Vereeniging utilizes scrap iron and pig iron from Natal. Hence statement 1 is not correct.**
- Iron ore is the source of primary iron for the world's iron and steel industries. Almost all (98%) iron ore is used in steelmaking. In Europe, Sweden is an important region where good quality iron ore is found. Major areas of iron-ore mining in Sweden include Kiruna, Gallivare, and Danmora. Another important region is Lorraine in France. **Lorraine has the largest iron-ore reserve in Europe. Apart from this,**

Normandy, Pyrenees, Selsia and Phalia region of Germany, and Cleveland, Midland and Scotland region of United Kingdom are major mining ore areas of Europe. Hence statement 2 is correct.

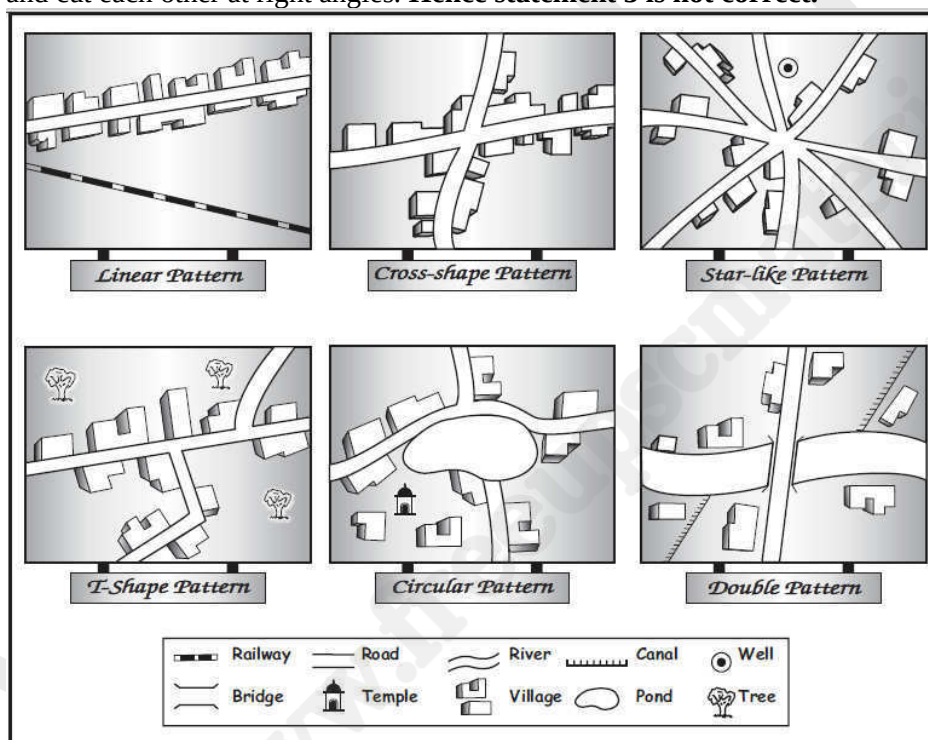
- The Ruhr region in western Germany is one of Europe's most densely populated regions and largest industrial centers. **Ruhr region is famous for coal storage and has high-quality coking coal.**

Q 31.D

- The term Mediterranean agriculture applies to the agriculture practiced in those regions which are having the Mediterranean type of climate. It is highly commercialized farming. **Hence statement 1 is correct.** There are five major regions in the world having the Mediterranean types of agriculture. The largest of these encircles is the Mediterranean Sea and it derives its name from this region. **Other regions include California in the USA, Central Chile, Southern part of South Africa. Hence statement 2 is not correct.**
- Viticulture or grape cultivation is a specialty of the Mediterranean region. **Hence statement 3 is correct.**
- The advantage of Mediterranean agriculture is that more valuable crops such as fruits and vegetables are grown in the winter season. It has great demands in Europe and America. **Hence statement 4 is correct.**

Q 32.C

- The different types of rural settlement patterns are:
 - **Circular** villages develop around lakes, and tanks, and sometimes the village is planned in such a way that the central part remains open and is used for keeping the animals to protect them from wild animals. **Hence statement 1 is not correct.**
 - **Star-like** patterns develop where several roads converge, star-shaped settlements develop by the houses built along the roads. **Hence statement 2 is correct.**
 - **Rectangular** patterns are found in plain areas or wide intermontane valleys. The roads are rectangular and cut each other at right angles. **Hence statement 3 is not correct.**



Q 33.D

- **Market gardening and horticulture** specialize in the cultivation of high-value crops such as vegetables, fruits, and flowers, solely for the urban markets.
 - **Farms are small and are located where there are good transportation links with the urban center where a high-income group of consumers is located.**
 - It is both labour and capital intensive and lays emphasis on the use of irrigation, HYV seeds, fertilizers, insecticides, greenhouses, and artificial heating in colder regions.
 - The regions where farmers specialise in vegetables only, the farming is known as truck farming. **The distance of truck farms from the market is governed by the distance that a truck can cover overnight, hence the name truck farming. Hence Pair 1 is correct.**
- In addition to market gardening, a modern development in the industrial regions of Western Europe and North America is **factory farming**.

- Livestock, particularly poultry and cattle rearing, is done in stalls and pens, fed on manufactured feedstuff and carefully supervised against diseases. **Hence Pair 2 is correct**
- This requires heavy capital investment in terms of building, machinery for various operations, veterinary services and heating and lighting. One of the important features of poultry farming and cattle rearing is breed selection and scientific breeding.
- **Mixed Farming is form of agriculture found in the highly developed parts of the world, e.g. North-western Europe, Eastern North America, parts of Eurasia and the temperate latitudes of Southern continents .**
 - Mixed farms are moderate in size and usually the crops associated with it are wheat, barley, oats, rye, maize, fodder and root crops.
 - Fodder crops are an important component of mixed farming. Crop rotation and intercropping play an important role in maintaining soil fertility. Equal emphasis is laid on crop cultivation and animal husbandry.
 - Animals like cattle, sheep, pigs and poultry provide the main income along with crops. **Hence Pair 3 is not correct.**
 - Mixed farming is characterised by high capital expenditure on farm machinery and building, extensive use of chemical fertilisers and green manures and also by the skill and expertise of the farmers.

Q 34.C

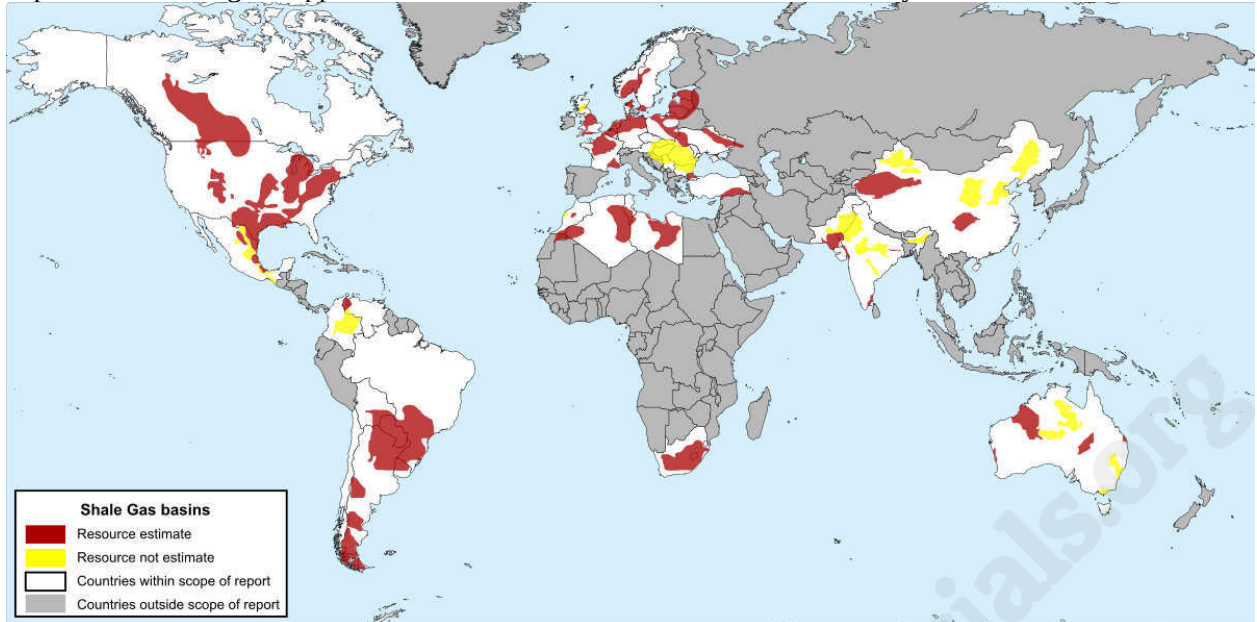


- **Golden Quadrilateral**
- It comprises the construction of a 5,846 km long 4/6 lane, high-density traffic corridor, to connect India's four big metro cities of Delhi-Mumbai-Chennai Kolkata.
- Other cities connected by this network include Ahmedabad, Bengaluru, Baleswar (Balasore), **Bhubaneswar**, Cuttack, Berhampur, Durgapur, **Jaipur**, Kanpur, Pune, Kolhapur, Surat, Vijayawada, Ajmer, Vizag, Bodhgaya, **Varanasi**, Allahabad, Agra, Mathura, Dhanbad, Gandhinagar, Udaipur, and Vadodara.
- **Hence option (c) is the correct answer.**

Q 35.A

- **Shale gas is a natural gas produced from shale, a type of sedimentary rock.** Shale gas has reduced America's dependence on oil imports, leading other countries to look for such reserves. **China is estimated to have the world's largest shale gas reserves followed by USA, Argentina, Mexico, South Africa, Australia, and Canada. United States has access to significant shale resources.** In the Annual Energy Outlook 2014, EIA (Energy Information Administration) estimated that the United States has approximately 610 Tcf of technically recoverable shale natural gas resources and 59 billion barrels of technically recoverable tight oil resources. **Hence option (a) is the correct answer.**
- **The United States and Canada are the major producers of commercially viable natural gas from shale formations in the world, even though about a dozen other countries have conducted exploratory test wells. China is the only nation outside of North America that has registered commercially viable production of shale gas, although the volumes contribute less than 1% of the total natural gas production in that country.**

- India, too, has potential to reduce its dependence on imports by tapping the potential of shale gas. **Six onshore basins — Cambay, Krishna-Godavari, Cauvery, Assam-Arakan, Ganga and Gondwana/Damodar—have been identified for shale exploration.**
- **Krishna Godavari basin, located in eastern India, is considered to hold the largest shale gas reserves in the country.** The basin is estimated to have around 27 Tcf of technically recoverable gas. The Cauvery basin in Tamil Nadu state is estimated to have recoverable shale gas reserves of 7 Tcf. The Cambay basin in Gujarat is the largest basin in the country, spread across 51,800 sq km.
- As per the initial studies, around 20 Tcf of gas is estimated as technically recoverable reserves in the Cambay basin. ONGC had drilled the country's first shale gas well in Jambusar in the October in 2013 to exploit the natural gas trapped within the shale formations located in Cambay basin.



Q 36.D

- **Shifting cultivation is a mode of farming long followed in the humid tropics of Sub-Saharan Africa, Southeast Asia, and South America.** In the practice of “slash and burn”, farmers would cut the native vegetation and burn it, then plant crops in the exposed, ash-fertilized soil for two or three seasons in succession.
- India has nearly 1.73 million hectares under shifting cultivation, where plots of land are cultivated temporarily and then abandoned for years so that the land becomes fertile again. An estimated 6.23 lakh families are involved in this activity, 90% of which are in the Northeast based on 2003 data.
- **It is known by different names in different parts:-**
 - It is jhumming in north-eastern states like Assam, Meghalaya, Mizoram, and Nagaland; Pamlou in **Manipur**, Dipa in Bastar district of **Chhattisgarh**, and in Andaman and Nicobar Islands.
 - **Jhumming:** The ‘slash and burn agriculture is known as ‘Milpa’ in Mexico and Central America, ‘Conuco’ in Venezuela, ‘Roca’ in Brazil, ‘Masole’ in Central Africa, ‘Ladang’ in Indonesia, ‘Ray’ in Vietnam.
 - In India, this primitive form of cultivation is called ‘**Bewar**’ or ‘**Dahiya**’ in **Madhya Pradesh**, ‘**Podu**’ or ‘**Penda**’ in **Andhra Pradesh**, ‘Pama Dabi’ or ‘Koman’ or Bringa’ in Odisha, ‘Kumari’ in the Western Ghats, ‘Valre’ or ‘Waltre’ in South-eastern Rajasthan, ‘Khil’ in the Himalayan belt, ‘Kuruwa’ in Jharkhand, and ‘Jhumming’ in the North-eastern region
- **Hence option (d) is the correct answer.**

Q 37.C

- **Depending upon the percentage of carbon present, the coal can be grouped into four types, such as peat, lignite, bituminous, and anthracite.**
 - **Peat-** It represents the first stage of coal formation, i.e. from wood to coal, today; peat is being formed at many places. It has a high percentage of moisture and volatile matter. The carbon content in peat is less than 40 percent. It burns like wood and gives more smoke and less heat. It leaves a large amount of ash after burning. Its low heating capacity reduces its value as an industrial fuel.
 - **Lignite-** It is generally regarded as the next stage of coal formation after peat. It is also known as brown coal. **Lignite is soft but more compact than peat.** The carbon content varies from 40 percent to 60 percent. Lignite has a large percentage of moisture and less amount of combustible matter. The

increasing demand for coal has enhanced its use in thermal power stations and in some industries. In India, lignite is mostly found in Rajasthan, Tamil Nadu, Assam, Jammu, and Kashmir.

- **Bituminous** -It is a very hard and compact variety of coal. The carbon content varies from about 60 percent to 80 percent. **Almost 80 percent of the world's total output of coal is of the bituminous type.** The moisture and the volatile contents are also less. Coke is mainly used in the iron and steel industry for smelting iron ore in blast furnaces. Bituminous coal is found in Jharkhand, Orissa, West Bengal, Madhya Pradesh, and Chhattisgarh.
- **Anthracite** - It is one of the best qualities of coal. The carbon content varies from about 80 percent to 90 percent. Anthracite, practically, has no volatile matter. It does not ignite easily, but once lighted, it has the highest heating capacity. It burns for a long time and leaves very little ash behind. It is harder and more lustrous than other types of coal. Hence statement 1 is correct.
- **Only about 5 percent of the world's total coal is anthracite.** In India, this type of coal is found only in Jammu and Kashmir, and that too in very small quantities. **Anthracite is the least plentiful form of coal. In the United States, it is found mostly in northeastern Pennsylvania and makes up less than 2 percent of all coal reserves in the country.** Smaller amounts of anthracite occur in South Africa, Australia, eastern Ukraine, western Canada, China, and other countries. **Hence statements 2 and 3 are not correct.**

Q 38.A

- **In agriculture, water is mainly used for irrigation.** Irrigation is needed because of Spatio-temporal variability in rainfall in the country.
- Provision of irrigation makes multiple cropping possible. It has also been found that irrigated lands have higher agricultural productivity than unirrigated land.
- **Further, the high-yielding varieties of crops need regular moisture supply, which is made possible only by developed irrigation systems.**
- Minor irrigation projects have both surface and groundwater as their source, while Major and Medium projects mostly exploit surface water resources. **Hence statement 1 is correct.**
 - Groundwater minor irrigation is primarily done through the individual and cooperative efforts of farmers with the help of institutional finance and their own savings.
 - Surface water minor irrigation schemes are generally funded from the public sector only.

Irrigation methods:

- Localized irrigation is a system where water is distributed under low pressure through a piped network, in a pre-determined pattern, and applied as a small discharge to each plant or adjacent to it. It is also known as a low-flow irrigation system/low-volume irrigation/micro-irrigation. **Drip irrigation, spray or micro-sprinkler irrigation, and bubbler irrigation belong to this category of irrigation methods. Hence statement 2 is not correct.**
- Furrow irrigation is conducted by creating small parallel channels along the field length in the direction of the predominant slope. Water is applied to the top end of each furrow and flows down the field under the influence of gravity.



Furrow irrigation using siphon tubes

Q 39.D

- Extensive Commercial grain cultivation is practiced in the interior parts of semi-arid lands of the midlatitudes. **Hence statement (a) is correct.**

- Wheat is the principal crop, though other crops like corn, barley, oats, and rye are also grown. **Hence statement (b) is correct.**
- The size of the farm is very large, therefore entire operations of cultivation from ploughing to harvesting are mechanized. **Hence statement (c) is correct.**
- There is a low yield per acre but a high yield per person. **Hence statement (d) is not correct.**
 - Efforts are always made to maximize per capita production in lieu of per unit land production. Each farmer controls and cultivates extensive farmland. **So per capita production increases significantly.**
 - **Due to the extensive nature of the cropping patterns, productivity per unit area of land remains very low.** Only in some parts of north-western Europe is productivity at par with intensive farming. In the rest of the extensive farming region, productivity often goes lower than intensive farming.
- This type of agriculture is best developed in Eurasian steppes, the Canadian and American Prairies, the Pampas of Argentina, the Velds of South Africa, the Australian Downs, and the Canterbury Plains of New Zealand.

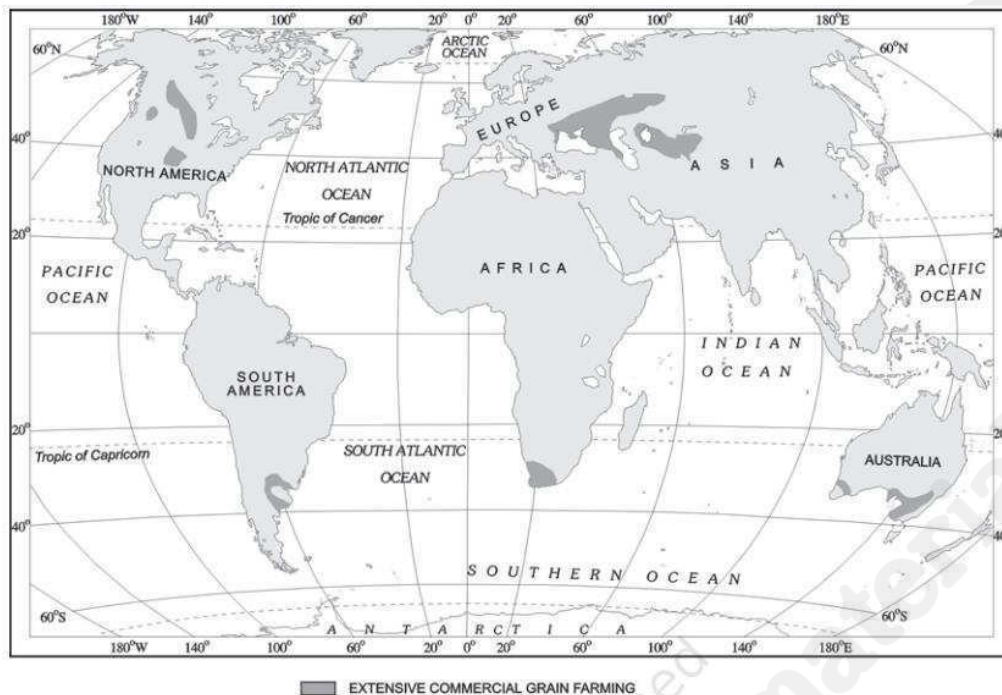


Fig. 5.12: Areas of Extensive Commercial Grain Farming

Q 40.A

- Footloose industry is a general term for an industry that can be placed and located at any location without effect from factors such as resources or transport.
- Footloose industries can be located in a wide variety of places. **They are not dependent on any specific raw material, weight losing, or otherwise. They largely depend on component parts which can be obtained anywhere. Hence statement 1 is correct.**
- The reason for the growth of footloose industries lies in the rapid development of highly sophisticated products requiring a great deal of scientific research and development. Footloose industries enable quick product improvement of their products to suit the market demand.
- Characteristic features of footloose Industries which favor the free choice of location:
 - light industries that often do not use raw materials but component parts
 - power requirements, usually only electricity — available from the national grid
 - the end product is small and often cheaper and easier to move;
 - employs a small labor force
 - **non-polluting industries which can be located near residential areas**
 - **accessibility - needs to be near a road network**
 - **Hence statements 2 and 3 are not correct.**

Q 41.B

- **Recent Context:** The Union Government has launched the 'SVANidhi se Samriddhi' program in additional 126 cities across 14 States/ UTs.
- **About the 'SVANidhi se Samriddhi':**
 - It is an additional program of the PM SVANidhi Scheme.

- Launched under the Ministry of Housing and Urban Affairs. **Hence statement 1 is not correct.**
- **Implementing Partner: Quality Council of India (QCI). Hence statement 3 is correct.**
- During Phase 1 of the program, it covered 125 cities, covering approximately 35 Lakh Street vendors and their families.
- Aim: To provide social security benefits to street vendors for their holistic development and socio-economic upliftment. **Hence statement 2 is not correct.**
- Under the program, socio-economic profiling of PM SVANidhi beneficiaries and their families is conducted to assess their eligibility for Eight Government of India welfare schemes and facilitate sanctions of eligible schemes.
- **PM SVANidhi Scheme:**
 - It is a special micro-credit facility plan to provide affordable loans of up to ₹10,000 to more than 50 lakh street vendors, who had their businesses operational on or before 24 March 2020.
 - **Small Industries Development Bank of India** is the technical partner for the implementation of this scheme.
 - It will manage the credit guarantee to the lending institutions through Credit Guarantee Fund Trust for Micro and Small Enterprises.

Q 42.D

- The population density is the concentration of the individuals within a specific geographic location. It is influenced by many factors.
- Physical factors such as favorable landforms like river valleys have a higher concentration of population. E.g. Ganga-Brahmaputra Plains in India. Fertile soil areas have high density across the globe. **Hence options 2 and 3 are correct.**
- Economic factors also influence population density. Regions with little or no economic opportunities tend to be sparsely populated. The industrial regions across the world are densely populated. Osaka in Japan and Mumbai in India are good examples. **Hence option 1 is correct.**
- Emigration is the outward movement of people from a particular area, it will lead to a reduction in population. **Hence option 4 is not correct.**
- **Immigration will lead to an increase in population density.**

Q 43.C

- Among the non-metallic minerals produced in India, mica is the important one.
- It has a unique combination of elasticity, toughness, flexibility, and transparency. It possesses resistance to heat and sudden changes in temperature and high dielectric strength. It is chemically inert, stable, and does not absorb water. **Mica is mainly used in the electrical and electronic industries. It can be split into very thin sheets which are tough and flexible. Hence statement 3 is correct.**
- Andhra Pradesh leads with a 41% share of the country's total resources followed by Rajasthan (28%), Odisha (17%), Maharashtra (13%), and Bihar (2%), and a small number of resources are found in Jharkhand and Telangana. **Hence statement 2 is not correct.**
- **In Rajasthan, the mica belt extends for about 320 km from Jaipur to Bhilwara and around Udaipur.**
- Mica deposits also occur in Mysuru and Hasan districts of Karnataka, Coimbatore, Tiruchirapalli, Madurai, and Kanniyakumari in Tamil Nadu, Alleppey in Kerala, Ratnagiri in Maharashtra, Purulia, and Bankura in West Bengal.
- As per the Government of India Notification S.O. 423(E), dated 10th February 2015, '**mica**' has been declared as '**Minor Mineral**'.
- It has also been used in India since ancient times as a medicinal item in Ayurveda and is known as Abhrak.
- Three major types of mica found in India are- Muscovite, Phlogopite, and Biotite.
- **India has a near-monopoly in the production of mica, producing about 60% of the world's total production. Hence statement 1 is correct.**
- Of late, there has been a steady downfall in the production of mica. This declining trend could be attributed to a falling in the demand for natural mica in the world market due to technological improvements that facilitate the use of reconstituted mica and the emergence of mica substitutes.

Production:

- Mica in India is produced in Jharkhand, Andhra Pradesh, Telangana, and Rajasthan followed by Tamil Nadu, West Bengal, and Madhya Pradesh.

- In Jharkhand, high-quality mica is obtained in a belt extending over a distance of about 150 km, in length and about 22 km, in width in the lower Hazaribagh plateau.
- In Andhra Pradesh, the Nellore district produces the best quality mica.

Q 44.D

- Manganese is a black hard iron-like metal and is an important raw material for the smelting of iron ore and is also used for manufacturing ferrous alloys. **Manganese is not found as a free element in nature and is often found in combination with iron.**
- **It is also used for the manufacture of bleaching powder, insecticides, paints, glazed pottery, matches, batteries, and china clay. Hence statement 1 is correct.**
- **South Africa is the world's largest producer of manganese**, accounting for 33.5% of global production. Its annual manganese production amounts to 6.2 million tonnes, with a majority of manganese mining concentrated in the Kalahari Desert, which is believed to hold more than 70% of global reserves. **With an annual production of about 1 million tonnes, Brazil is the fifth-largest manganese producer in the world. Brazil has 10% of the world's manganese reserves, behind Ukraine (24%), South Africa (22%), and Australia (16%). Hence statement 2 is not correct.**
- Four-fifths of the total production is consumed domestically and exports constantly decreasing due to increasing domestic demand. **Japan is the largest buyer of Indian manganese accounting for about two-thirds of the total export.** The other buyers are the USA, UK, Germany, France, and Norway. **Hence statement 3 is correct.**

Q 45.D

- The development of the iron and steel industry opened the doors to rapid industrial development in India. Almost all sectors of the Indian industry depend heavily on the iron and steel industry for their basic infrastructure. **The other raw materials besides iron ore and coking coal, essential for the iron and steel industry are limestone, dolomite, manganese, and fire clay. Therefore all four raw materials are used in the manufacturing process.**
- All these raw materials are gross (weight loss), therefore, the best location for the iron and steel plants is near the source of raw materials. **In India, there is a crescent-shaped region comprising parts of Chhattisgarh, Northern Orissa, Jharkhand, and western West Bengal, which is extremely rich in high-grade iron ore, good quality coking coal, and other supplementing raw materials.**
- **Hence option (d) is the correct answer.**

Q 46.C

- The Europeans colonized many parts of the world and they introduced some other forms of agriculture such as plantations which were mainly profit-oriented large-scale production systems and known as commercial farming.
 - The main aim of commercial farming is to produce crops as per the market demands. It can be either intensive or extensive. To keep the **cost of production low**, most modern methods of cultivation are employed.
 - **It is generally practiced in areas of a sparse population.**
 - **In India, commercial farming is not very common due to the heavy pressure of the population on land.** Commercial agriculture has developed in Punjab, Haryana, Gujarat, Maharashtra, Uttar Pradesh, West Bengal, Assam
 - **The main characteristic of this type of farming is the use of higher doses of modern inputs**, e.g. high yielding variety (HYV) seeds, chemical fertilizers, insecticides, and pesticides in order to obtain higher productivity. The degree of commercialization of agriculture varies from one region to another. For example, rice is a commercial crop in Haryana and Punjab, but in Odisha, it is a subsistence crop.
- **Subsistence farming** categories and types are only for personal use and not for commercial usage or selling in the market. This farming technique henceforth doesn't require heavy machinery or technology. This is the most common type of traditional farming.
- **Nomadic farming** is more of a method where the farmers or cultivators do not stick permanently to a place, instead keep moving with animals in search of water and land, pasture. This practice and technique are heavily present in arid to semi-arid regions.
- Unlike just following on rearing animals or entirely focusing on crops, the types of **mixed farming** incorporate crop cultivation and livestock in a single farm.
- **Hence option (c) is the correct answer.**

Q 47.B

- **Recent Context:** Punjab will be the first state in the country to provide "digitized form J" in real-time to farmers from this Rabi procurement season.
- 'J form' is the sale receipt of a farmer's agricultural produce in mandis (grain market) which acts as income proof for a farmer who sells his crop.
 - These forms were earlier issued manually by arthiyas (commission agents).
 - It can be used for raising finance, IT waivers, subsidy claims, farmers' insurance, etc.
 - It will act as a record of land under cultivation for both the wheat and paddy crops in the state.
 - It can also be stored in DigiLocker.
- **Hence option (b) is the correct answer.**

Q 48.C

- **Recent Context:** The President of Turkey approved the changes in Electoral Laws to bolster his rule.
- Turkey is a large peninsula that bridges the continents of Europe (via. Eastern Thrace) and Asia (via. the Anatolian Plateau).
- **Political Boundaries:**
 - Bulgaria and Greece in the Northwest;
 - Russia, Ukraine, and Romania are connected through the Black Sea;
 - Georgia, Armenia, Azerbaijan, and Iran to the east; and
 - Syria and Iraq to the South.
- **Geographical Features:**
 - Dardanelles, the Sea of Marmara, and the Bosphorus Strait divide the country between Europe and Asia while connecting the Black Sea with the Mediterranean Sea (via the Aegean Sea).
 - Largest lake: Lake Van
 - Main Rivers: Euphrates, Kizilirmak and Sakarya
 - Highest Peak: Mt Ararat
- **Hence option (c) is the correct answer.**



Q 49.D

- **Copper is an indispensable metal in the electrical industry for making wires, electric motors, transformers, and generators.** Copper has been used for making utensils and coins for a long. It is alloyable, malleable, and ductile. It is also mixed with gold to provide strength to jewellery. India has limited reserves of copper.
- India's refined copper imports more than doubled in 2018-19 to 92,290 tonnes (\$605.20 million) from 44,245 tonnes (\$294.95 million) in 2017-18. Export of refined copper, on the other hand, has substantially dropped to 47,971 tonnes (\$302.27 million) in 2018-19 from 378,555 tonnes (\$2,435.57 million) in 2017-18. So much so, the country has now become a net importer of refined copper. India has to import copper from other countries, mainly the USA, Canada, Zimbabwe, and Mexico. **Hence statement 1 is not correct**
- The Copper deposits mainly occur in the Singhbhum district in Jharkhand, Balaghat district in Madhya Pradesh, and Jhunjhunu and Alwar districts in Rajasthan. **Hence statement 2 is not correct**
- Minor producers of Copper are Agnigundala in Guntur District (Andhra Pradesh), Chitradurga and Hasan districts (Karnataka), and South Arcot district (Tamil Nadu).
- Mining copper is a costly and tedious affair as copper ores contain a small percentage of metal. **Hence statement 3 is correct**
- **The largest reserves/resources of copper ore to the tune of 813 million tonnes (53.81%) are in the State of Rajasthan** followed by Jharkhand with 295 million tonnes (19.54%) and Madhya Pradesh with 283 million tonnes (18.75%). Copper reserves/ resources in Andhra Pradesh, Gujarat, Haryana, Karnataka, Maharashtra, Meghalaya, Nagaland, Odisha, Sikkim, Tamil Nadu, Telangana, Uttarakhand, and West Bengal accounted for the remaining 7.9% of the total All India resources.
- **Madhya Pradesh was the leading producer State of copper concentrates accounting for about 53% of the production during 2018-19, followed by Rajasthan with 42% and Jharkhand with 4 percent.** Malanjkhand Copper Project [M.P] is the largest copper ore producing mine with a 2.0 million tonnes production capacity per year.



Q 50.D

- **A recent report said that India will likely miss its 2022 target of installing 100 gigawatts (GW) of solar power capacity.** As of April 2022, only about 50% of the 100GW target, consisting of 60GW of utility-scale and 40GW of rooftop solar capacity, has been met.
- **India's Solar sector:**
 - Since 2011, India's solar sector has grown at a compounded annual growth rate (CAGR) of around 59% from 0.5GW in 2011 to 55GW in 2021. **Hence statement 2 is correct.**
 - **India currently ranks fifth in terms of installed solar power capacity.** **Hence statement 1 is not correct.**

- Causes for lag:
 - The disruption in supply chains due to the pandemic.
 - Limits to net-metering (or paying users who give back surplus electricity to the grid).
 - Taxes on imported cells and modules, etc.
- Key recommendations:
 - Consistent regulations for net metering and banking facilities to apply nationally,
 - Stricter enforcement of the renewable purchase obligation (RPO) by states,
 - A capital subsidy for battery energy storage systems (BESS), etc.
- **India has set a target of achieving a total installed capacity of 100GW by 2022 and 300GW by 2030 under the Jawaharlal Nehru National Solar Mission/National Solar Mission (launched in 2010). Hence statement 3 is correct.**

Q 51.B

- A wide range of low-land tropical cash crops is cultivated in a tropical monsoon climate. The most important crop in this category is cane sugar as much as two-thirds of the world's sugar production comes from tropical countries. Sugar is either grown in plantations or on smallholdings whenever rainfall and sunshine are abundant. **Hence, pair 1 is not correctly matched.**
- The olive tree is extensively cultivated in the Mediterranean climate. It is hardy and long-rooted and can survive even on very poor limestone soils with less than 10 inches of annual rainfall
- Cocoa is a tropical plantation crop that is well suitable in a hot and wet equatorial climate. It is most extensively cultivated in west Africa bordering of Gulf of Guinea. The two most important producers are Ghana and Nigeria. **Hence pairs 2 and 3 are correctly matched.**

Q 52.D

- Crude oil is a type of fossil fuel which occurs naturally. It is an unrefined petroleum product composed of hydrocarbon deposits and other organic materials. **Crude oil is classified into two types, Sweet crude and Sour Crude depending on the content of Sulphur present in it.**
- **Crude oil may be referred to as sweet if it contains relatively little sulfur (0.5%) or sour if it contains substantial amounts of sulfur. Hence statement 1 is not correct.**
- **Sweet crude requires less energy to be extracted and once extracted, yields higher quality gasoline as well as larger quantities of it. Hence statement 2 is correct.**
- **Iraq is one of the leading producers of sweet crude.** Major locations where sweet crude is found include the Appalachian Basin in Eastern North America, Western Texas, the Bakken Formation of North Dakota and Saskatchewan, the North Sea of Europe, North Africa, Australia, and the Far East including Indonesia. **Hence statement 3 is not correct.**
- Sour crude, on the other hand, has a high level of impurities in it, namely sulfur, which must first be removed before being processed into gas and other petroleum based products. **Venezuela is a leading producer of sour crude oil.**

Q 53.D

- **Collective farming or Kolkhoz farming** was introduced by the Soviet Union. **Hence statement 1 is not correct.**
- The basic principle behind this method was social ownership of means of production and collective labor.
- It was introduced to improve on the previous model of farming and improve efficiency and self-sustainability in agricultural production. In cooperative farming, wherein farmers pool their resources and ownership remains as per their share in resources so there is no social ownership. So, Kolkhoz was not about cooperative farming. **Hence statement 2 is not correct.**

Q 54.C

- Pulses are a very important ingredient of vegetarian food as these are rich sources of proteins.
- India is the largest producer and consumer of pulses in the world, contributing nearly 24 percent to the global output. **Hence statement (b) is correct.**
- **India is also the largest importer of pulses. Rajasthan, Maharashtra, and Madhya Pradesh are the major pulse-producing state in the country. Hence statement (d) is correct.**
- A large number of pulses are grown across multiple agro-climatic zones:
 - Kharif: Arhar (pigeonpea); urad (black gram); moong (green gram); Barbati (cowpea), lathyrus
 - Rabi: gram; pea; ramjash masoor (lentil)
 - Zaid (summer): urad; moong; cow pea
- The average yield of all pulses is about 660 kg/hectare compared to the world average of 909 kg/hectare.

- Surprisingly, the average yield of Beharin is 18,485 kg/hectare, the highest in the world. It is obvious that there is a vast scope to increase the yield.
- The cultivation of pulses in the country is largely concentrated in the drylands of Deccan and central plateaus and northwestern parts of the country.
- Pulses occupy about 11 percent of the total cropped area in the country. Being the rainfed crops of drylands, the yields of pulses are low and fluctuate from year to year. **Hence statement (a) is correct.**
- Pulses need less moisture and survive even in dry conditions. **Being leguminous crops, all these crops except arhar help in restoring soil fertility by fixing nitrogen from the air. Therefore, these are mostly grown in rotation with other crops. Hence statement (c) is not correct.**

Table 5.2(b): Distribution of Area under major Crops

(Figures in Percentage)

Year	Rice	Wheat	Nutri Cereals	Pulses	Foodgrains	Oilseeds	Sugarcane	Cotton	Others	All Crops
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
2007-08	23.31	14.88	15.12	12.54	65.85	14.17	2.68	5.00	12.30	100.00
2008-09	24.17	14.73	14.57	11.73	65.19	14.63	2.34	4.99	12.85	100.00
2009-10	22.59	15.33	14.91	12.55	65.38	13.99	2.25	5.46	12.93	100.00
2010-11	21.85	14.82	14.44	13.46	64.56	13.88	2.49	5.73	13.35	100.00
2011-12	22.48	15.25	13.50	12.50	63.72	13.44	2.57	6.22	14.05	100.00
2012-13	22.13	15.52	12.82	12.04	62.51	13.71	2.59	6.20	15.00	100.00
2013-14	22.07	15.24	12.61	12.61	62.54	14.03	2.50	5.98	14.95	100.00
2014-15	22.29	15.90	12.72	11.90	62.80	12.93	2.56	6.48	15.23	100.00
2015-16	22.10	15.45	12.39	12.66	62.60	13.25	2.50	6.24	15.40	100.00
2016-17	22.19	15.53	12.62	14.85	65.19	13.21	2.24	5.46	13.90	100.00
2017-18	22.07	14.95	12.25	15.03	64.31	12.36	2.39	6.35	14.60	100.00
2018-19	23.01	15.28	11.54	15.19	65.01	12.92	2.64	6.57	12.86	100.00
2019-20	22.06	15.85	12.11	14.28	64.30	13.62	2.30	6.74	13.04	100.00

Source: Directorate of Economics & Statistics

Notes: (1) Area estimates are based on data provided by State Statistical Authorities (SASAs) which is further cross checked and validated with estimates provided by other agencies, viz., MNCFC and CWWG.

(2) Others include Jute & Mesta, Coconut, major Horticulture crops, Tea, Coffee and Rubber

(3) Data for Foodgrains, Oilseeds and Commercial Crops are as per 4th Advance Estimates for 2019-20

(4) Data for Horticulture crops are as per 3rd Advance Estimates for 2019-20

Q 55.D

- **The Bindibu or Aborigines is a tribe of Australia.** They are primitive hunters and collectors. They inhabit the Great Sandy Desert and Gibson Desert of Australia. **Hence pair 1 is not correctly matched.**
- **The Bushmen are the tribe of the Kalahari Desert in the African continent.** They are primitive hunters and collectors. **Hence pair 2 is correctly matched.**
- **The Gobi Mongols are nomadic herdsmen.** Hence pair 3 is not correctly matched.

Q 56.D

- **Recent Context:** There are no reports to indicate that the Indian tent turtle is on the verge of extinction due to illegal mining in Narmada River, according to the Union Minister of State for Environment, Forest & Climate Change. He further informed that the Zoological Survey has not conducted any survey in the Narmada river on the impact of illegal mining on Indian tent turtles and its effect on the river ecosystem.
- **Indian Tent Turtle (Pangshura Tentoria) is a Schedule-I species in India under the Wild Life (Protection) Act, 1972. Hence statement 1 is correct.**
- **IUCN Status: Least Concern. Hence statement 2 is correct.**
- It is native to India, Nepal, and Bangladesh and is found in freshwater rivers, swamps, and ponds. Its habitats include still water pools on the riverside and slow running water near the river banks. They are active swimmers and mainly herbivorous.
 - **The common habitats in India include - Orissa, Maharashtra, Andhra Pradesh, Assam, Madhya Pradesh, etc. Hence statement 3 is not correct.**



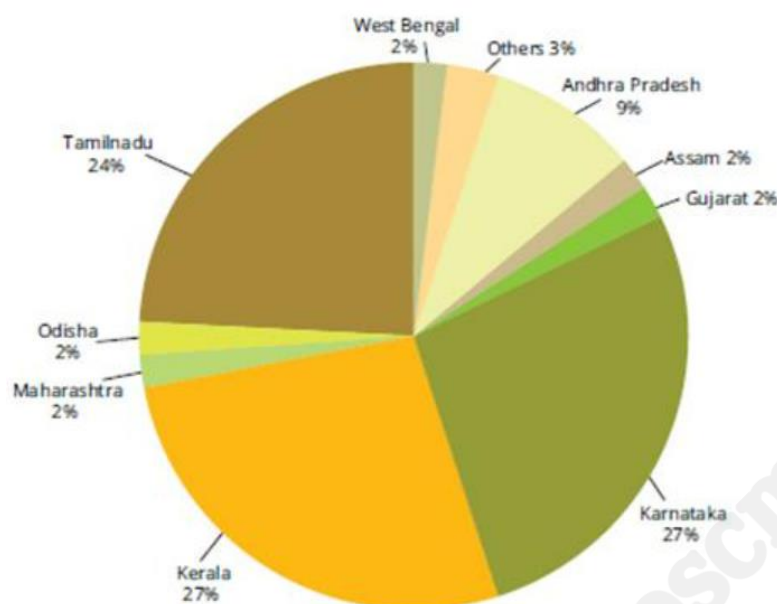
- The State of **Rajasthan** lies between **23.3 to 30.12 North latitude** and 69.30 to 78.17 East longitude.
- The state of **Odisha** extends from 17.31N latitude to **22.31N latitude**.
- **Madhya Pradesh** is located in the geographic heart of India between the latitude of 21.6°N–26.30°N and longitude of 74°9'E–82°48'E.
- **Gujarat** state is located between **20° to 24°40' N latitude**.
- Therefore when the **northernmost point of states** is arranged from north to south Rajasthan comes first then Madhya Pradesh, then Gujarat, and then Odisha.
- Hence option (a) is the correct answer.

Q 58.A

- **Plantation** is also a type of commercial farming. In this type of farming, a single crop is grown in a large area. Hence statement 3 is not correct.
- The plantation has an **interface between agriculture and industry**. Plantations cover large tracts of land, using capital-intensive inputs, with the help of migrant laborers. All the produce is used as raw material in respective industries. In India, tea, coffee, rubber, sugarcane, banana, etc., are important plantation crops. Tea in Assam and North Bengal coffee in Karnataka are some of the important plantation crops grown in these states.
- **Plantation agriculture was introduced by the Europeans** in colonies situated in the tropics. Some of the important plantation crops are tea, coffee, cocoa, rubber, cotton, oil palm, sugarcane, bananas, and pineapples.

- The characteristic features of this type of farming are large estates or plantations, large capital investment, managerial and technical support, scientific methods of cultivation, single crop specialization, cheap labor, and a good system of transportation that links the estates to the factories and markets for the export of the products.
 - The French established cocoa and coffee plantations in west Africa.
 - The British set up large tea gardens in India and Sri Lanka, rubber plantations in Malaysia, and sugarcane and banana plantations in West Indies.
 - Spanish and Americans invested heavily in coconut and sugarcane plantations in the Philippines. The Dutch once had a monopoly over sugarcane plantations in Indonesia.
 - **Some coffee fazendas (large plantations) in Brazil are still managed by Europeans.** Today, ownership of the majority of plantations has passed into the hands of the government or the nationals of the countries concerned. **Hence statement 2 is correct.**
- Extensive Commercial Grain Cultivation is best developed in Eurasian steppes, the Canadian and American Prairies, the Pampas of Argentina, the Velds of South Africa, the Australian Downs, and the Canterbury Plains of New Zealand. **Hence statement 1 is not correct.**

Leading Plantation Crops Producing States



Q 59.B

- The agro-climatic classification is a climate classification keeping in view the suitability for agriculture.
- National commission on agriculture (1971) classified the country into 127 agro-climatic zones. **Hence statement 1 is not correct.**
- The planning commission, as a result of mid. term appraisal of planning targets of VII plan (1985 - 90) divided the country into 15 broad agro-climatic zones based on physiographic and climate. **Hence statement 2 is correct.**
- The emphasis was given on the development of resources and their optimum utilization in a suitable manner within the framework of resource constraints and potentials of each region.

The 15 agro-climatic zones are:-

Zone 1 - Western Himalayan Region: Jammu and Kashmir, Uttar Pradesh

Zone 2 - Eastern Himalayan Region: Assam, Sikkim, West Bengal and all North-Eastern states

Zone 3 - Lower Gangetic Plains Region: West Bengal

Zone 4 - Middle Gangetic Plains Region: Uttar Pradesh, Bihar

Zone 5 - Upper Gangetic Plains Region: Uttar Pradesh

Zone 6 - Trans-Gangetic Plains Region: Punjab, Haryana, Delhi and Rajasthan

Zone 7 - Eastern Plateau and Hills Region: Maharastra, Uttar Pradesh, Orissa and West Bengal

Zone 8 - Central Plateau and Hills Region: MP, Rajasthan, Uttar Pradesh

Zone 9 - Western Plateau and Hills Region: Maharastra, Madhya Pradesh and Rajasthan

Zone 10 - Southern Plateau and Hills Region: Andhra Pradesh, Karnataka, Tamil Nadu

Zone 11 - East Coast Plains and Hills Region: Orissa, Andhra Pradesh, Tamil Nadu and Pondicherry

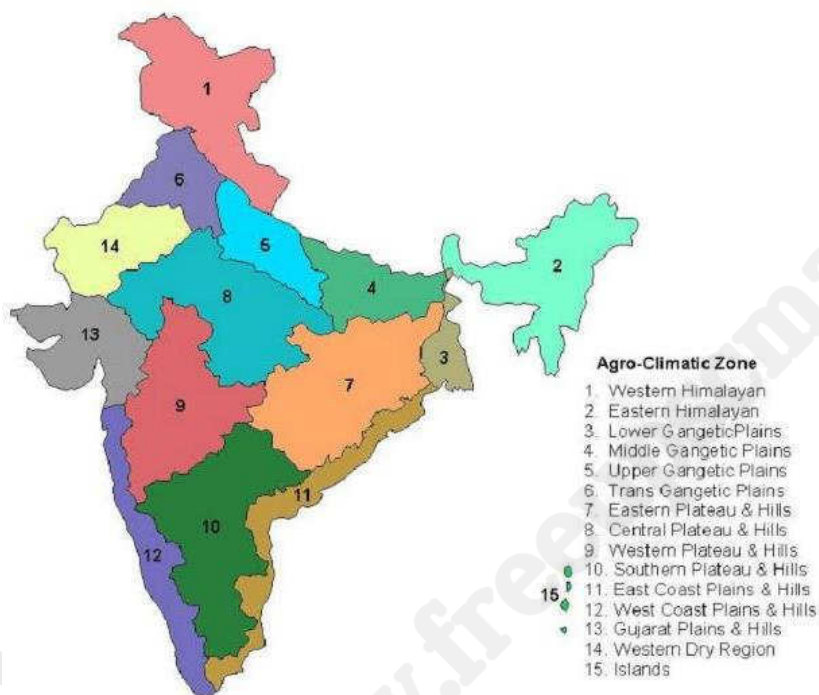
Zone 12 - West Coast Plains and Ghat Region: Tamil Nadu, Kerala, Goa, Karnataka, Maharashtra

Zone 13 - Gujarat Plains and Hills Region: Gujarat

Zone 14 - Western Dry Region: Rajasthan

Zone 15 - The Islands Region: Andaman and Nicobar, Lakshadweep. (Map 1)

Agro-climatic zones of India



Q 60.A

- A mineral is a natural substance of organic or inorganic origin with definite chemical and physical properties. India is endowed with a rich variety of mineral resources due to its varied geological structure.
- Bulk of the valuable minerals are **products** of pre-Palaeozoic and are mainly associated with metamorphic and igneous rocks of peninsular India. The vast alluvial plain tract of north India is devoid of minerals of economic use.
- Metallic minerals are the sources of metals. Iron ore, copper, and gold produce metal and are included in this category.
- **Metallic minerals are further divided into ferrous and non-ferrous metallic minerals.**
 - **Ferrous, refers to iron.** All those minerals which have iron content are ferrous such as iron ore itself and those which do not have iron content are non-ferrous such as copper, bauxite, etc.
 - **The non-ferrous metals industry constitutes several sub-sectors –**
 - Base metals (aluminium, copper, zinc, lead, nickel, tin)

- Precious metals (silver, gold, palladium, other platinum group metals)
- Minor metals including refractory metals (e.g. tungsten, molybdenum, tantalum, niobium, chromium) and
- Specialty metals (e.g. cobalt, germanium, indium, tellurium, antimony, and gallium).
- Non-metallic minerals are either organic in origins such as fossil fuels also known as mineral fuels which are derived from buried animal and plant life such as coal and petroleum.
- Other types of non-metallic minerals are inorganic in origin such as mica, limestone, graphite, etc.
- **Hence option (a) is correct.**
- **Minerals have certain characteristics:**
 - These are unevenly distributed over space.
 - There is an inverse relationship between quality and quantity of minerals i.e. good quality minerals are less in quantity as compared to low-quality minerals.
 - The third main characteristic is that all minerals are exhaustible over time. These take long to develop geologically and they cannot be replenished immediately at the time of need.

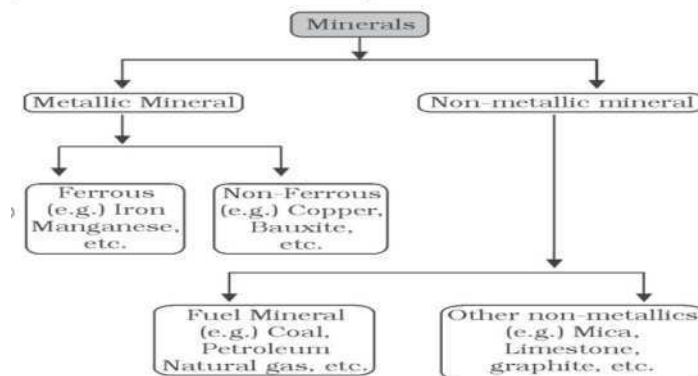


Fig. 7.1 : Classification of Minerals

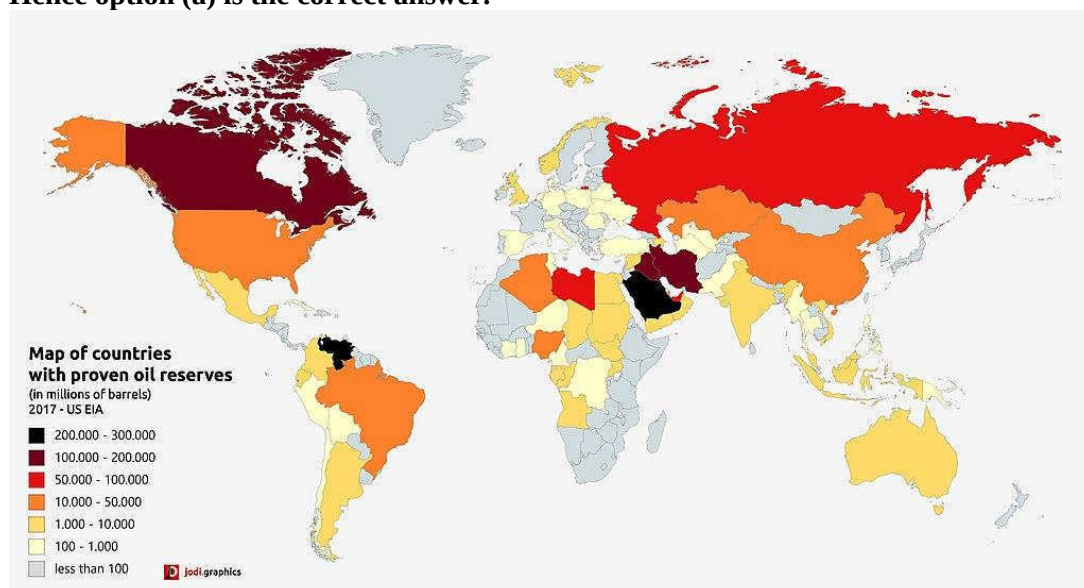
Q 61.C

- Both growth and development refer to changes over a period of time.
- The difference is that growth is quantitative and value neutral. It may be either positive (showing an increase) or negative (indicating a decrease).
- Development means a qualitative change which is always value positive. This means that development cannot take place unless there is an increment or addition to the existing conditions. **Hence, statement 1 is correct but statement 2 is not correct.**
- Development occurs when positive growth takes place. Yet, positive growth does not always lead to development.
- For example, if the population of a city grows from one crore to two crore over a period of time, we say the city has grown. However, if facilities like housing, provision of basic services and other characteristics remain the same, then this growth has not been accompanied by development. **Hence, statement 3 is not correct.**

Q 62.A

- Petroleum is not distributed evenly around the world. **More than half of the world's proven oil reserves are located in the Middle East.** Following the Middle East are Canada and the United States, Latin America, Africa, and the region occupied by the former Soviet Union. Each of those regions contains less than 15 percent of the world's proven reserves.
- The nearly 280 world-class giant fields thus far discovered, plus the super-giants, account for about 80 percent of the world's known recoverable oil. There are, in addition, approximately 1,000 known large oil fields that initially contained between 50 million and 500 million barrels. These fields account for some 14 to 16 percent of the world's known oil. Major oil fields are listed below:
 - **Ghawar field – Saudi Arabia**
 - **Burgan field – Kuwait**
 - Azeri-Chirag-Guneshli – Caspian Sea, Azerbaijan
 - Ku-Maloob-Zaap – Mexico
 - Zakum – UAE
 - Ferdows field – Iran
 - **Sugar Loaf field – Brazil**
 - Bolivar Coastal field – Venezuela

- World's five largest offshore oilfields:
 - Safaniya oilfield – Persian Gulf, Saudi Arabia
 - Upper Zakum oilfield – Persian Gulf, UAE
 - Manifa oilfield – Persian Gulf, Saudi Arabia
 - **Kashgan oilfield – Caspian Sea, Kazakhstan**
 - Lula Oilfield – Brazil
- Hence option (a) is the correct answer.



Q 63.B

- **Recent Context:** The Rathotsava or chariot festival began at Karnataka's Chennakeshava Temple with the traditional recital of the Quran even as Hindu groups sought the discontinuation of the ritual which they believe is against their religion.
- **The Chennakeshava Temple (also called Vijaya Narayana Temple), was built by Hoysala King Vishnuvardhana to commemorate his victory over the Cholas in 1116 AD.** The temple is believed to have been built by master craftsmen – Dasoja and Chavana, a father and son duo.
- **It is located at Belur in Karnataka (also known earlier as Velapuri, Velur, and Belapur in olden times) and is situated on the banks of the Yagachi River and was one of the capitals of the Hoysala Empire.**
- This **star-shaped temple** is believed to have taken around 103 years to build.
 - At the **entrance facing the temple is a towering Gopura and a magnificent sculpture of Garuda**, Lord Vishnu's carrier, palms touching in pious homage.
 - The temple stands on a platform and has exquisite artwork on its outer walls adorned with **bracket figures depicting the Puranas and Epics, all carved to perfection.**
 - The lower frieze has a **series of charging elephants and horses**; the elephants symbolize courage while the horses are for speed.
 - It is also believed that Shantalaidevi, the queen of King Vishnuvardhana is the model for one of the sculptures – Darpana Sundari (lady with the mirror).
 - Just outside is an interesting piece of sculpture, called the **Gravity pillar which is carved out of a single rock and stands in its own weight.**
 - Equally impressive are the **temples of Kappe Chennigaraya, Soumyanayaki, Andal, and other Vaishnava manifestations, which surround the main temple.**
 - The **sanctum has a magnificent 3.7 m tall image of Lord Vijaya Narayana in black stone.** The prabhavali of this idol has the 10 avatars of Lord Vishnu, wrought very perfectly.
 - The doorway with the dwarapalakas is very elegantly carved.
 - **The Garbhagriha is stellar in shape and its zigzag walls make the figures of 24 forms of Lord Vishnu look different at different times of the day due to light and shade effect.**
 - One can see the lathe-turned pillars highly polished and intricate carvings even in the dark and no two pillars are alike. The Narasimha Pillar, it seems, was built to rotate on its base. The Mohini Pillar is the most beautiful among the pillars. A unique aspect in this is the artist has left a small space blank – it is believed that it is a challenge to others to do a better carving and to convey that art never dies as it is infinite.
- **The temple is also proposed to be listed under UNESCO World Heritage Sites.**

Q 64.A

- Mineral fuels are essential for the generation of power, required by agriculture, industry, transport, and other sectors of the economy.
- Energy can be generated from fuel minerals like coal, petroleum, natural gas, uranium, and electricity. Energy resources can be classified as conventional and non-conventional sources.
 - **Conventional sources include** firewood, cattle dung cake, coal, petroleum, natural gas, and electricity (both hydel and thermal).
 - **Non-conventional sources include** solar, wind, tidal, geothermal, biogas, and atomic energy
- **Depending upon its source and utilization, energy can be divided into two major classifications :**
 - **Traditional or non-commercial Energy:** Examples of non-commercial energy resources are firewood, charcoal, cow dung, and agricultural wastes.
 - **Commercial Energy:** The commercial sources of energy comprise coal, oil, natural gas, hydroelectricity, natural gas, nuclear power as well as wind and solar power.
- **Hence option (a) is the correct answer.**

Q 65.A

- A natural gas field is a deposit rich in natural gas. Nearly 80% of the world's natural gas fields are located in ten nations. **Russia's reserves alone account for almost a quarter of the total global gas reserves and also hold two of the largest fields: Urengoy and Yamburg. Shtokman field is also one of the world's largest natural gas fields, which lies in the Barents Sea, north Russia.**
- **Three of the major Siberian fields, namely Yamburg, Urengoy, and Medvezh'ye, account for approximately 45% of the country's gas reserves.** The majority of the country's gas reserves under development and production are located in the Nadym-Pur-Taz (NPT) region of upper Western Siberia.
- Russia holds 1,688 trillion cubic feet (Tcf) of proven gas reserves as of 2017, ranking 1st in the world and accounting for about 24% of the world's total natural gas reserves of 6,923 Tcf.
- **Hence option (a) is the correct answer.**



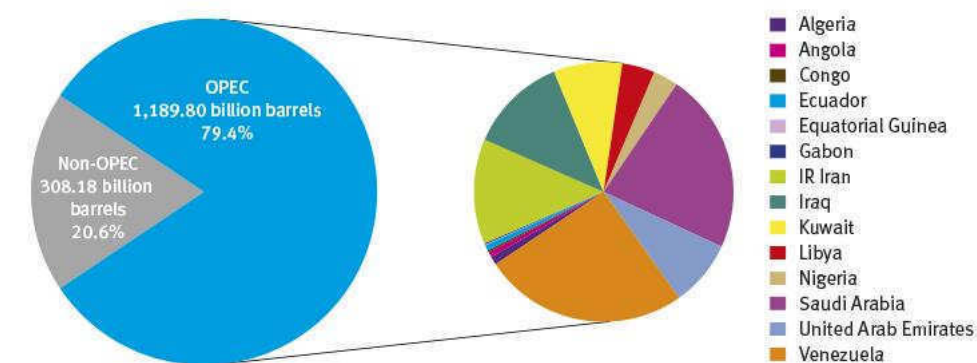
Q 66.D

- Oil is a natural resource formed by the decay of organic matter over millions of years. And like many other natural resources, oil cannot be produced, only extracted where it already exists. Unlike every other natural resource, oil is the lifeblood of the global economy. **Global oil reserves are not distributed evenly across the globe. More than half of the world's proven oil reserves are located in the Middle East. Following the Middle East are Canada and the United States, Latin America, Africa, and the region occupied by the former Soviet Union. Each of those regions contains less than 15 percent of the world's proven reserves. Hence statement 1 is correct.**
- **Venezuela has the largest amount of oil reserves in the world with more than 300 billion barrels in reserve. Saudi Arabia has the second-largest amount of oil reserves in the world with 297.5 billion barrels.** Although Venezuela has the most oil reserves in the world, most of its oil is offshore or far underground and is considered to be dense. As such, the cost of extracting the oil in Venezuela's reserves

using the technology currently available is too high to be profitable. Saudi Arabia's oil reserves, on the other hand, are close to the surface and on land, which makes the oil much more accessible and the extraction process vastly more cost-effective. This, in turn, makes Saudi Arabia's oil sector significantly more profitable. **Hence statement 2 is not correct.**

- **The Ghawar oilfield, located about 100km southwest of Dhahran in the Al Hasa Province of Saudi Arabia, is the world's biggest conventional oil field both by oil reserves and production.** The Ghawar field currently produces about 3.8 million barrels of oil and 2.5 billion cubic feet of natural gas per day. **It is by far the biggest oil-producing field and is one of the world's six fields producing more than one million barrels of oil per day. Ghawar is estimated to account for about 6% of the world's total daily crude oil output.** The field's production peaked at 5.7 million barrels per day in 1981 and later slipped below the five million. **Hence statement 3 is not correct.**

OPEC share of world crude oil reserves, 2018



OPEC proven crude oil reserves, at end 2018 (billion barrels, OPEC share)

Venezuela	302.81	25.5%	Kuwait	101.50	8.5%	Algeria	12.20	1.0%	Gabon	2.00	0.2%
Saudi Arabia	267.03	22.4%	UAE	97.80	8.2%	Ecuador	8.27	0.7%	Equatorial Guinea	1.10	0.1%
IR Iran	155.60	13.1%	Libya	48.36	4.1%	Angola	8.16	0.7%			
Iraq	145.02	12.2%	Nigeria	36.97	3.1%	Congo	2.98	0.3%			

Source: OPEC Annual Statistical Bulletin 2019.

Q 67.B

- **Recent Context:** Government has announced that nearly 2,71,000 persons will be trained over five years under the PM-DAKSH scheme.
- **Pradhan Mantri Dakshta Aur Kushalta Sampann Hitgrahi (PM-DAKSH) yojana** is a skill development scheme from **Ministry of Social Justice and Empowerment**. **Hence statement 1 is not correct.**
 - Starting from 2020-21, it aims to provide free training to persons from SCs/OBCs, Safai Karamcharis, etc. through **Up-skilling/Reskilling**, Short Term Training Programme (focus on self-employment), **Long Term Training Programme**, and **Entrepreneurship Development Program**. **Hence statement 3 is not correct.**
 - **Eligibility:**
 - Marginalized persons of SC (Scheduled Caste), OBC (Other Backward Classes), Economically Backward Classes, Denotified tribes, Sanitation workers including waste pickers, manual scavengers, transgenders, and other similar categories.
 - **Implementation is through 3 Corporations: (hence statement 2 is correct).**
 - National Scheduled Castes Finance and Development Corporation
 - National Backward Classes Finance & Development Corporation
 - National Safai Karamcharis Finance & Development Corporation

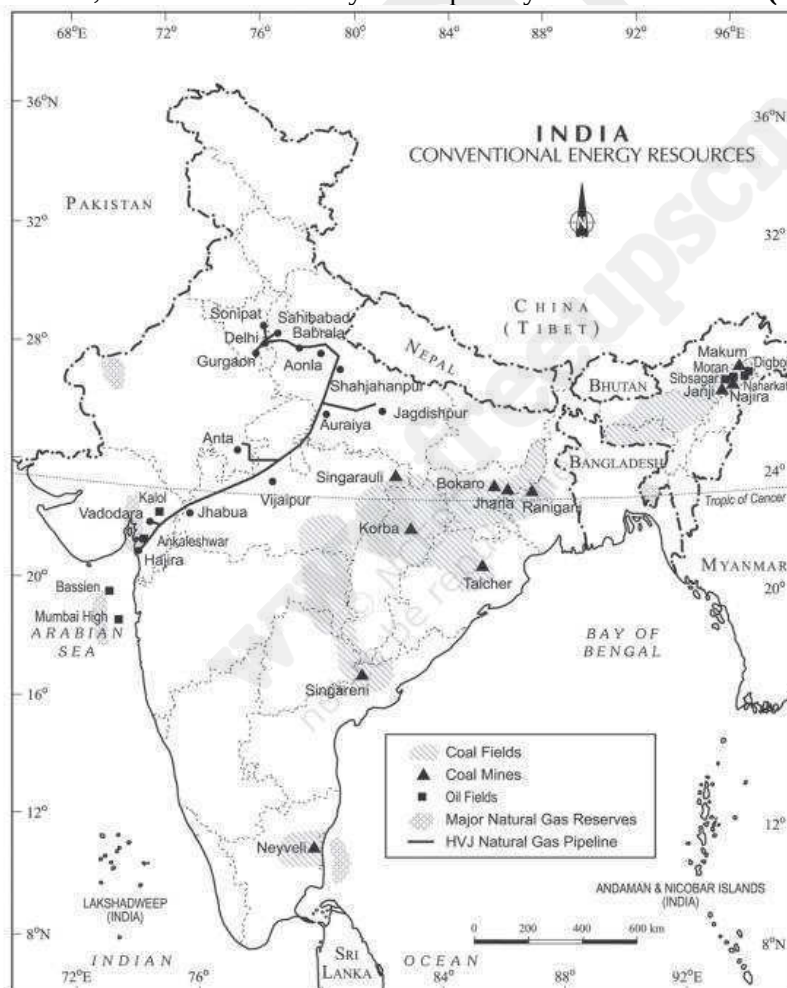
Q 68.D

- Coal occurs in rock series of two main geological ages, namely Gondwana and tertiary deposits. **Hence statement (d) is not correct.**
- About 65 percent of the total coal production is consumed for generating electricity. The entire process of steel making is based on metallurgical coal. The cement industry consumes about five percent of the total production. Coal is also an important source of naphtha and ammonia, which are widely used for making chemical fertilizers, tar, benzene, carbon black, etc. Soft coke is used as fuel in the kitchen.
- **India, in 2018 ranked 2nd in the world coal production.**
- **Chhattisgarh is the largest coal-producing state with a share of about 22.2% followed by Odisha with a contribution of 19.8% to the national output.**

- **The most important Gondwana coal fields** of India are located in Damodar Valley. They lie in the Jharkhand-Bengal coal belt and the important coalfields in this region are Raniganj, Jharia, Bokaro, Giridih, and Karanpura. Jharia is the largest coalfield followed by Raniganj. **Hence statement (b) is correct.**
- The other river valleys associated with coal are the Godavari, Mahanadi, and Sone. The most important coal mining centers are Singrauli in Madhya Pradesh (part of Singrauli coalfield lies in Uttar Pradesh), Korba in Chhattisgarh, Talcher, and Rampur in Odisha, Chanda-Wardha, Kamptee, and Bander in Maharashtra, and Singareni in Telangana and Pandur in Andhra Pradesh.
- **Tertiary coals** occur in Assam, Arunachal Pradesh, Meghalaya, and Nagaland. It is extracted from Darangiri, Cherrapunji, Mewlong, and Langrin (Meghalaya); Makum, Jaipur, and Nazira in upper Assam, Namchik – Namphuk (Arunachal Pradesh), and Kalakot (Jammu and Kashmir).

Depending upon the percentage of carbon present, the coal can be grouped into four types:

- **Peat**- It represents the first stage of coal formation, i.e. from wood to coal, today; peat is being formed in many places. It has a high percentage of moisture and volatile matter. The carbon content in peat is less than 40 percent. It burns like wood and gives more smoke and less heat. It leaves a large amount of ash after burning. Its low heating capacity reduces its value as an industrial fuel.
- **Lignite**- It is generally regarded as the next stage of a coal formation after peat. It is also known as brown coal. Lignite is soft but more compact than peat. The carbon contents vary from 40 percent to 60 percent. Lignite has a large percentage of moisture and less amount of combustible matter.
- **Bituminous** -It is the hard and compact variety of coal. The carbon content varies from about 60 percent to 80 percent. Almost 80 percent of the world's total output of coal is of the bituminous type. The moisture and the volatile contents are also less. Coke is mainly used in the iron and steel industry for smelting iron ore in blast furnaces. About 80 percent of the coal deposits in India are of a bituminous type and are of non-coking grade. **Hence statement (c) is correct.**
- **Anthracite** - It is the hardest and the best quality of coal. The carbon content varies from about 80 percent to 90 percent. Anthracite, practically, has no volatile matter. It does not ignite easily, but once lighted, it has the highest heating capacity. It burns for a long time and leaves very little ash behind. Only about 5 percent of the world's total coal is anthracite. In India, this type of coal is found only in Jammu and Kashmir, and that too is in very small quantity. **Hence statement (a) is correct.**



World Coal Reserves and Producers:

- The USA has the largest coal reserves with about 24% share of the total world reserves, followed by the Russian Federation (15%), Australia (14%), and China (13%)
- China continued to be the largest producer of coal & lignite in 2018 with about 46% share in total world production, followed by India (9.3%), the USA (8.6%), Australia (6.3%), and Indonesia (7%).

Q 69.B

- **Petroleum is also called 'black gold' or 'liquid gold'.** It is second to coal in terms of sources of energy. It is formed when large quantities of dead organisms, usually zooplankton and algae, are buried underneath sedimentary rock and subjected to intense heat and pressure. **Crude petroleum occurs in sedimentary rocks of the tertiary period. Hence statement 1 is correct.**
- **Hydraulic fracturing, or fracking, is a drilling method used to extract petroleum (oil) or natural gas from deep in the Earth.** In the fracking process, cracks in and below the Earth's surface are opened and widened by injecting water, chemicals, and sand at high pressure. Hydraulic fracturing can occur during three stages of the petroleum development cycle – exploration, evaluation and production. **Hence statement 2 is correct.**
- As of 31 March 2019, India had estimated crude oil reserves of 618.95 million tonnes (MT), increasing by 4.1% from the previous year. **Geographical distribution of crude oil indicates that the maximum reserves are in the western offshore including Bombay High and Rajasthan (52%) followed by Assam (23%) whereas maximum reserves of natural gas are in the Eastern offshore including CBM in West Bengal (38%) followed by western offshore including Bombay High, Rajasthan, Madhya Pradesh and Jharkhand (36%). Hence statement 3 is not correct.**

Q 70.C

- Primitive subsistence agriculture or shifting agriculture is being practiced by many tribes in the topics, such as central America, South East Asia, and Africa. It is known by different names in different regions. **It is known as Jhumming in India. Hence pair 1 is not correctly matched.**
- It is known as Milpa in Central America and Mexico. It is known as Ladang in Indonesia and Malaysia. **Hence pair 2 is correctly matched and pair 3 is not correctly matched.**

Q 71.D

- **Rhine Waterway** - The Rhine flows through industrially advanced nations of Switzerland, Germany, France, Belgium, and the Netherlands. The Rhine River is navigable from Rotterdam for about 870 km. At its source in Switzerland, it flows along the boundary of France and Germany and drains Germany and the Netherlands, and has its mouth near the North Sea. **Hence option (d) is the correct answer.**
- On its banks are located main cities of Europe like Strasbourg {France}, Bonn, Cologne, Dusseldorf, and Rotterdam. The vessel takes the cargo of industrial products, and coal food grains in addition to passengers and tourists which have seen a considerable rise in the past few decades. For the benefit of tourists, the vessels are fitted with modern conveniences and sail in both directions of the rift valley. One can enjoy the scenic beauty of the Vossages of France and the Black Forest of Germany. Each year more than 20,000 ocean-going ships and 2, 00,000 inland vessels sail in this waterway.



- **River Danube:** The Danube is the second-longest river in Europe, after the Volga in Russia. It flows through much of Central and Southeastern Europe, from the Black Forest into the Black Sea.
- **River Mississippi:** The Mississippi River is the second-longest river and chief river of the second-largest drainage system on the North American continent, second only to the Hudson Bay drainage system. From its traditional source of Lake Itasca in northern Minnesota, it flows generally south for 2,340 miles (3,770 km) to the Mississippi River Delta in the Gulf of Mexico.
- **River Volga:** The Volga is the longest river in Europe. Situated in Russia, it flows through Central Russia to Southern Russia and into the Caspian Sea.

Q 72.D

- **Recent Context: Kuwait's PM resigned as the political crisis intensifies.**



- Hence option (d) is the correct answer.

Q 73.A

- **Coal bed methane (CBM) is an unconventional form of natural gas found in coal deposits or coal seams. CBM is formed during the process of coalification, the transformation of plant material into coal.** It is especially used in industries such as in cement, rolling mills, steel plants, and methanol production. In recent decades it has become an important source of energy in the United States, Canada, Australia, and other countries. **India has the fifth-largest proven coal reserves in the world. Hence statement 1 is correct.**
- CBM can be found almost anywhere there is coal. **Global CBM production totals 5.8 Bcfd (billion cubic feet per day) from 15 basins in the USA, Canada, Australia, China, and India. The United States remains the world leader in CBM exploration, booked reserves, and production. Currently, there is commercial coalbed gas production or exploration in approximately 15 US basins. However, activity has slowed substantially in response to low gas prices but the USA still dominates with nearly 5Bcfd of production and about 20 Tcf produced to date.** The major producing areas are the Powder River, San Juan, Black Warrior, Central Appalachian, Raton, and Uinta basins. Other United States areas with significant exploration or production are the Cherokee, Arkoma, Illinois, Hanna, Gulf Coast, and Greater Green River basins. CBM production in China (150 MMcfd) and India (10 MMcfd) is struggling due to more challenging geologic conditions and low good productivity. **Hence statement 2 is not correct.**

Q 74.B

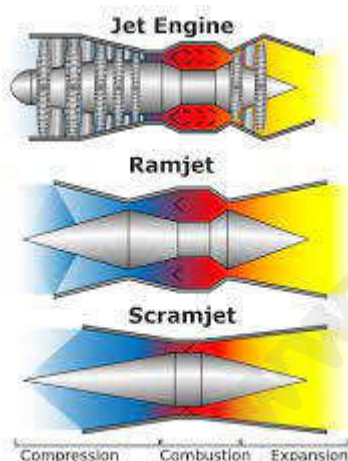
- The coarse cereals together occupy about 16.50 percent of the total cropped area in the country. Among these, jowar or sorghum alone accounts for about 5.3 percent of the total cropped area. It is the main food crop in semi-arid areas of central and southern India.
- **Maharashtra alone produces more than half of the total jowar production of the country. Hence statement 1 is not correct.**
- Other leading producer states of jowar are Karnataka, Madhya Pradesh, Andhra Pradesh, and Telangana. It is sown in both Kharif and rabi seasons in southern states. But it is a Kharif crop in northern India.

where it is mostly grown as a fodder crop. South of Vindhyachal, it is a rainfed crop and its yield level is very low in this region. **Hence statement 2 is correct.**

- Jowar is grown both as a Kharif and as a rabi crop. As a Kharif crop, it grows in areas having a mean monthly average temperature of 26°C to 33°C. It requires rainfall of about 30cm during the growing season.
- **Jowar can be grown in a variety of soils including loamy and sandy soils. The clayey, regur, and alluvium are most suitable for Jowar cultivation.**
- Bajra is sown in hot and dry climatic conditions in the northwestern and western parts of the country. It is a hardy crop that resists frequent dry spells and drought in this region.
- It is grown in areas with rainfall of about 40-50 cm and temperatures of about 25°C -30°C.
- **Bajra grows well on sandy soils and shallow black soil. Hence statement 3 is correct.**
- **Leading producers of bajra are the states of Maharashtra, Gujarat, Uttar Pradesh, Rajasthan, and Haryana.** Being a rainfed crop, the yield level of this crop is low in Rajasthan and fluctuates a lot from year to year.

Q 75.D

- **Recently, DRDO successfully flight-tests Solid Fuel Ducted Ramjet (SFDR) technology.**
- A ram jet is a form of air-breathing jet engine that uses the vehicle's forward motion to compress incoming air for combustion **without a rotating compressor. Hence statement 1 is not correct.**
 - Fuel is injected into the combustion chamber where it mixes with the hot compressed air and ignites.
 - **A ramjet-powered vehicle requires an assisted take-off** like a rocket assist to accelerate it to a speed where it begins to produce thrust.
 - **Ramjets work most efficiently at supersonic speeds around Mach 3** (three times the speed of sound) and can operate up to speeds of Mach 6. However, the ramjet efficiency starts to drop when the vehicle reaches hypersonic speeds. **Hence statement 2 is not correct.**
- **A scramjet engine** is an improvement over the ramjet engine as it efficiently operates at hypersonic speeds and allows supersonic combustion. Thus it is known as Supersonic Combustion Ramjet or Scramjet.
- **A dual-mode ramjet (DMRJ)** is a type of jet engine where a ramjet transforms into a scramjet over the Mach 4-8 range, which means it can efficiently operate both in subsonic and supersonic combustor modes.
- **SFDR is a missile propulsion system** that includes a thrust modulated ducted rocket with a reduced smoke nozzle-less missile booster.
 - It utilizes a **solid fuelled air-breathing Ramjet Engine**. Unlike solid-propellant rockets, the Ramjet takes up oxygen from the atmosphere during flight. Thus, it is light in weight and can carry more fuel.
 - As per **DRDO, SFDR-based propulsion enables** the missile to intercept aerial threats at very long range at supersonic speeds. SFDR helps to develop long-range air-to-air missiles.



Q 76.A

- The **Human Development Index (HDI)** is a summary measure of average achievement in key dimensions of human development:
 - a long and healthy life,
 - being knowledgeable and
 - have a decent standard of living.

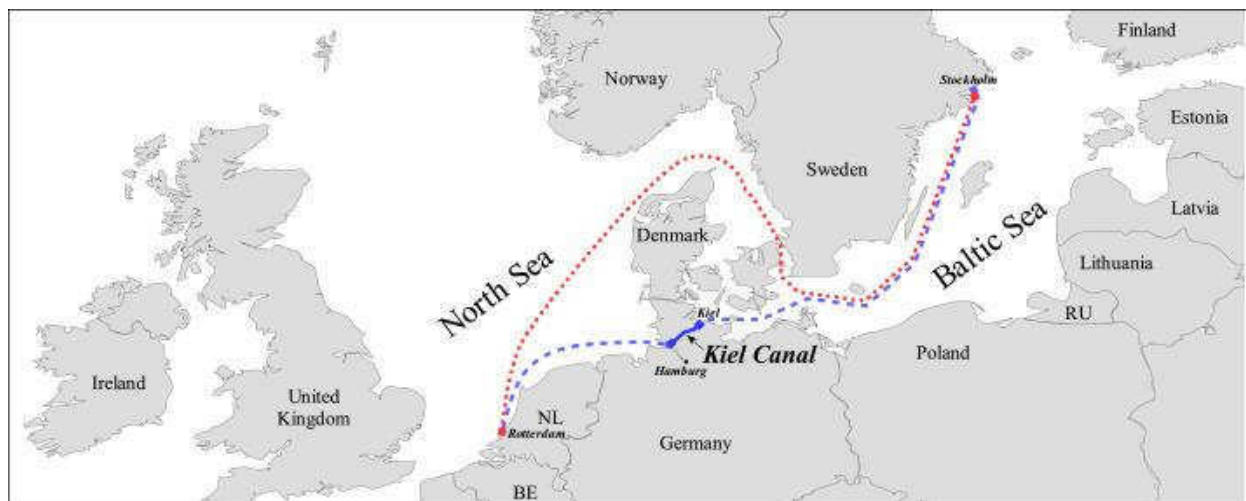
- The HDI is the geometric mean of normalized indices for each of the three dimensions.
- The health dimension is assessed by **life expectancy at birth**, the education dimension is measured by the **mean of years of schooling for adults aged 25 years and more and expected years of schooling for children of school entering age**. The standard of living dimension is measured by **gross national income per capita**. Therefore **GDP data is not used for measuring HDI**.
- Hence option (a) is the correct answer.

Q 77.A

- **Recent Context:** Space Weather Prediction Centre (SPWC) under National Oceanic and Atmospheric Administration (NOAA) has issued two geomagnetic storms (GMS) watches.
- GMS is **strong disturbance to Earth's magnetic field** and poses problems for many activities, technological systems, and critical infrastructure. **Hence statement 1 is correct.**
 - o **When Coronal Mass Ejections (CME) collide with the Earth, it causes GMS.**
The Earth's magnetic field changes in the course of a storm as the near-Earth system attempts to adjust to the jolt of energy from the Sun carried in the solar wind. Their effects can disturb the geomagnetic field for days at a time.
The most visible attribute of a geomagnetic storm is the aurora which becomes brighter and moves closer to the equator.
 - o Geomagnetic storms usually last a few hours to days. The strongest storms may persist for up to a week.
 - o A string of CMEs may cause prolonged disturbed periods related to the additional energy being pumped into Earth's magnetic field.
 - o The frequency of geomagnetic storms, in general, depends on where Earth is in the solar cycle—with most storms occurring near solar maximum; however, these storms are also common in the declining phase due to highspeed solar wind streams.
 - o Geomagnetic storms induce currents that can have a significant impact on electrical transmission equipment. Electric power companies have procedures in place to mitigate the impact of geomagnetic storms.
GMS is defined by changes in the Dst (disturbance – storm time) index. The Dst index estimates the globally averaged change of the horizontal component of the Earth's magnetic field at the magnetic equator based on measurements from a few magnetometer stations. Geomagnetic storm intensity is reported in several different ways, including, K-index, A-index, and the G-scale. **Hence statement 2 is not correct.**
- There are many kinds of eruptions of the sun. Solar flares and coronal mass ejections both **involve gigantic explosions of energy but are otherwise quite different**. The two phenomena do sometimes occur at the same time – indeed the strongest flares are almost always correlated with coronal mass ejections when the motion of the sun's interior contorts its own magnetic fields. Like the sudden release of a twisted rubber band, **the magnetic fields explosively realign, driving vast amounts of energy into space. This phenomenon can create a sudden flash of light - a solar flare.**
- **Flares can last minutes to hours and they contain tremendous amounts of energy.** Traveling at the speed of light, it takes eight minutes for the light from a solar flare to reach Earth.
 - o **The magnetic contortions can also create a different kind of explosion that hurls solar matter into space. These are the coronal mass ejections, also known as CMEs.**
The flare is like the muzzle flash, which can be seen anywhere in the vicinity. The CME is like the cannonball, propelled forward in a single, preferential direction, this mass ejected from the barrel only affects a targeted area. This is the CME—an immense cloud of magnetized particles hurled into space.

Q 78.A

- **The Kiel Canal is an important waterway in northern Germany, one of the most traveled man-made shipping lanes worldwide.** The Kiel Canal took more than 9000 workers eight years to build in 1887. Initially served German military needs by eliminating the necessity for ships to travel northward around Jutland (peninsular Denmark). **Hence option (a) is the correct answer.**
- **It opened in 1895 and mainly shortens the passage between the North Sea and the Baltic Sea.** The canal constitutes the safest, most convenient, shortest, and cheapest shipping route between the two seas.



Q 79.B

- The **Rourkela Steel plant** was set up in 1959 in the Sundargarh district of Orissa in collaboration with Germany. The plant was located on the basis of proximity to raw materials, thus, minimizing the cost of transporting weight losing raw materials.
- **Rourkela Steel plant** has a unique locational advantage, as it **receives coal from Jharia (Jharkhand) and iron ore from Sundargarh and Kendujhar**. The Hirakud project supplies power for the electric furnaces and water is obtained from the Koel and Sankh rivers. **Hence, pair 1 is not correctly matched.**
- The **Bhilai Steel Plant** was established with Russian collaboration in the Durg district of Chhattisgarh and started production in 1959. **The iron ore comes from the Dalli-Rajhara mine, coal comes from Korba and Kargali coal fields**. The water comes from the Tanduladam and the power from the Korba Thermal Power Station. This plant also lies on the Kolkata-Mumbai railway route. The bulk of the steel produced goes to the Hindustan Shipyard at Vishakhapatnam. **Hence, pair 2 is correctly matched.**
- **Durgapur Steel Plant in West Bengal** was set up in collaboration with the government of the United Kingdom and started production in 1962. This plant **lies in Raniganj and Jharia coal belt and gets iron ore from Noamundi**. Durgapur lies on the main Kolkata-Delhi railway route. Hydel power and water are obtained from the Damodar Valley Corporation (DVC). **Hence, pair 3 is not correctly matched.**

Q 80.D

- India accounts for about 2.45 percent of the world's surface area, 4 percent of the world's water resources, and about 16 percent of the world's population. The availability of surface water and replenishable groundwater is 1,869 cubic km. **Out of this, only 60 percent can be put to beneficial uses.**
- **There are four major sources of surface water. These are rivers, lakes, ponds, and tanks.** The amount of water that can actually be put to beneficial use is much less due to severe limitations imposed by physiographic, topographic, interstate issues, and the present technology to harness water resources economically.
- India has a vast coastline and the coast is very indented in some states. Due to this, a number of lagoons and lakes have formed. States like Kerala, Odisha, and West Bengal have vast surface water resources in these lagoons and lakes. **Although water is generally brackish in these water bodies, it is used for fishing and irrigating certain varieties of paddy crops, coconut, etc. Hence statement 1 is not correct.**
- Agriculture accounts for most of the surface and groundwater utilization, it accounts for 89 percent of the surface water and 92 percent of the groundwater utilization. **While the share of the industrial sector is limited to 2 percent of the surface water utilization and 5 percent of the groundwater, the share of the domestic sector is higher (9 percent) in surface water utilization as compared to groundwater. Hence statement 2 is not correct.**
- The share of the agricultural sector in total water utilization is much higher than other sectors. However, in the future, with development, the shares of industrial and domestic sectors in the country are likely to increase.

Q 81.D

- **Recent Context:** Centre would establish Seaweed park with Special Economic Zone status in Tamil Nadu to improve the livelihood of fishermen.
 - It will be implemented under the Prime Minister's Matsya Sampada Yojana.

- Seaweeds, are the primitive, marine **non-flowering** marine algae **without** root, stem and leaves, plays a major role in marine ecosystems. **Hence option (d) is the correct answer.**
 - Range in colours from red, green, brown and black and also vary in size, from microscopic to large underwater forests.
 - Large seaweeds form dense underwater forests known as **kelp forests**, which act as underwater nurseries for fish, snails and sea urchins. On their part, the seaweeds derive nutrition through photosynthesis of sunlight and nutrients present in seawater. **They release oxygen through every part of their bodies.**
 - Found mostly in intertidal region (Southern Gulf of Mannar), shallow and deep waters of the sea and estuaries and backwaters.
 - **Significance of seaweeds**
 - Storehouse of nutrients like iron, zinc, magnesium, riboflavin, thiamine, vitamin A, B, C & K etc.
 - **Act as a bio-indicator** and balance out the ecosystem. When waste from agriculture, industries, aquaculture and households are let into the ocean, it causes nutrient imbalance leading to algal blooming, the sign of marine chemical damage.
 - ✓ **These aquatic organisms heavily rely on iron for photosynthesis.** When quantity of this mineral exceeds healthy levels and becomes dangerous to marine life, seaweeds trap it and prevent damage. **Similarly, most heavy metals found in marine ecosystems are trapped and removed by seaweeds.**
 - ✓ Mitigating climate change.
 - ✓ Used as fertilizers and to increase fish production.
 - ✓ Can be buried in beach dunes to combat beach erosion.
 - ✓ Used as an ingredient in preparing toothpaste, cosmetics and paints.

Q 82.B

- **Recently, RBI released guidelines on the establishment of Digital Banking Units (DBUs).**
- DBU is a specialized fixed point business unit/hub housing a certain minimum infrastructure for delivering digital banking products and services to ensure convenient and cost-effective banking for customers.
- DBUs aim to enhance the spread of digital modes of banking in India.
- **Major guidelines:**
 - All domestic scheduled commercial Banks (**excluding Regional Rural Banks, Payments Banks, and Local Area Banks**) with past digital banking experience are permitted to open DBUs.
 - Products and services provided at a DBU include **opening of accounts, cash withdrawal and deposit, KYC updation, loans, and complaint registrations.**
 - **Each DBU will be treated as Banking Outlets** and each DBU needs to be housed distinctly, with the separate entry and exit provisions. **Hence option (b) is the correct answer.**
 - Adequate safeguards for the cyber security of the DBUs will be ensured by the banks.
 - An adequate digital mechanism offers real-time assistance and redresses customer grievances arising from business and services offered by the DBUs.
- **Significance:**
 - Helpful for farmers and senior citizens in rural areas, enabling interoperability and financial inclusion.
 - Provide a boost to the digital economy.
 - Reduce physical footprint with fewer brick and mortar branches.
 - Cheaper to establish than a new branch, require lesser staff, with cheaper maintenance due to technological tools and hence can be high-yield units for the parent bank.

Q 83.C

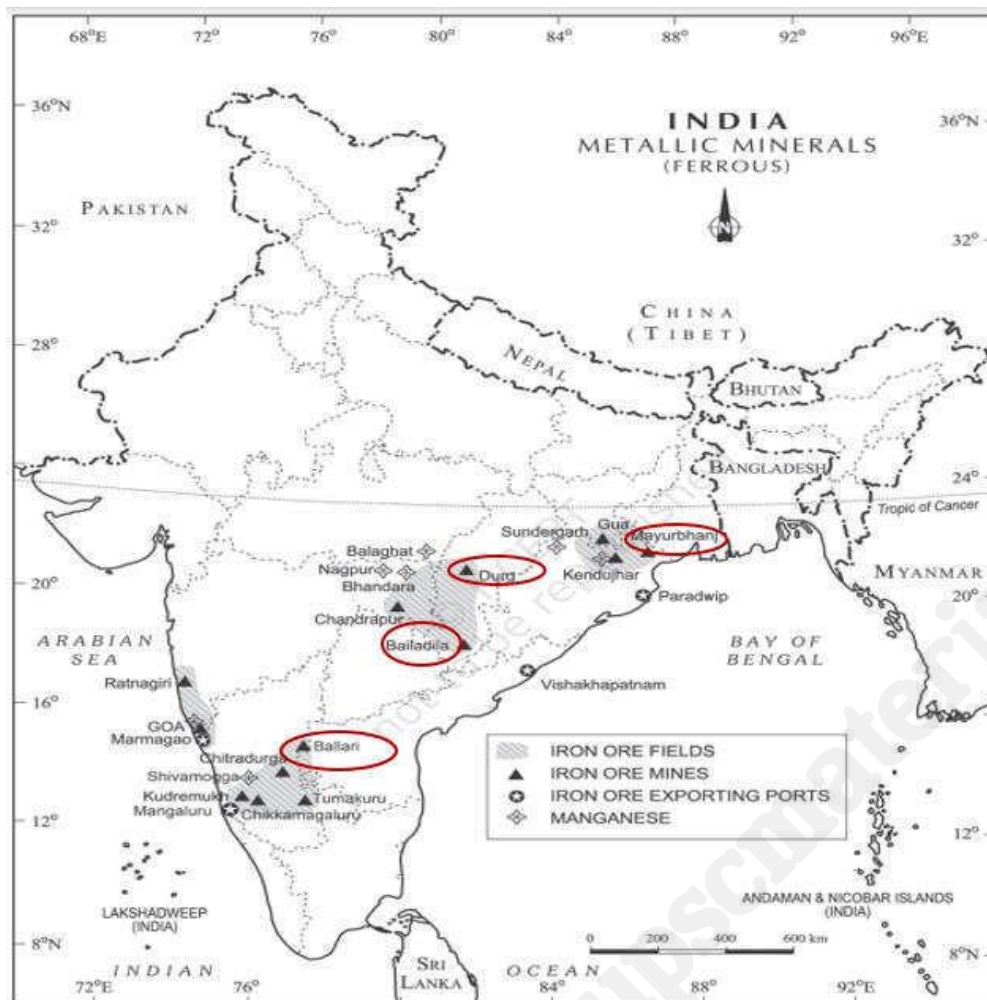
- Iron & steel is the driving force behind industrial development in any country.
- **Haematite and magnetite are the most important iron ores in India.** About 79% of haematite ore deposits are found in the Eastern Sector (Assam, Bihar, Chhattisgarh, Jharkhand, Odisha & Uttar Pradesh) while about 93% of magnetite ore deposits occur in Southern Sector (Andhra Pradesh, Goa, Karnataka, Kerala & Tamil Nadu). **Karnataka alone contributes 72% of magnetite deposits in India**

LOCATION OF IRON ORE MINES:-

- **In Odisha**, iron ore occurs in a series of hill ranges in Sundergarh, Mayurbhanj, and Jhar. The important mines are Gurumahisani, Sulaipet, Badampahar (Mayurbhaji), Kiruburu (Kendujhar) and Bonai (Sundergarh).
- Similar hill ranges, **Jharkhand** has some of the oldest iron ore mines and most of the iron and steel plants are located around them. Most of the important mines such as Noamundi and Gua are located in Poorbi

and Pashchimi Singhbhum districts. This belt further extends to Durg, Dantewara, and Bailadila. Dalli and Rajhara in Durg are the important mines of iron ore in the country.

- In **Karnataka**, iron ore deposits occur in the Sandur -Hospet area of Ballari district, Baba Budan hills, and Kudremukh in Chikkamagaluru district and parts of Shivamogga, Chitradurg, and Tumakuru districts.
- The districts of Chandrapur, Bhandara, and Ratnagiri in Maharashtra, Karimnagar, and Warangal district of Telangana, Kurnool, Cuddapah, and Anantapur districts of Andhra Pradesh, Salem and Nilgiris districts of Tamil Nadu are other iron mining regions. Goa has also emerged as an important producer of iron ore.
- **Hence, option (c) is the correct answer.**



Q 84.D

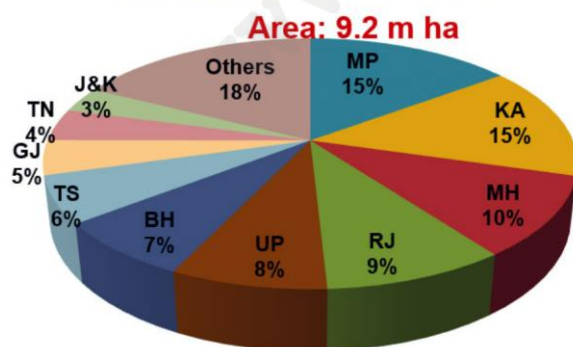
- **Clustered Settlements**
- The **clustered rural settlement is a compact or closely built-up area of houses**. In this type of village, the general living area is distinct and separated from the surrounding farms, barns, and pastures.
- The closely built-up area and its intervening streets present some recognizable pattern or geometric shape, such as rectangular, radial, linear, etc.
- Such settlements are generally found in fertile alluvial plains and in the northeastern states. Sometimes, **people live in compact villages for security or defense reasons, such as in the Bundelkhand region of central India and in Nagaland**. In Rajasthan, scarcity of water has necessitated compact settlement for maximum utilization of available water resources. **Hence statement 1 is correct.**
- **Semi-Clustered Settlements**
- **Semi-clustered or fragmented settlements** may result from the tendency of clustering in a restricted area of dispersed settlement. More often such a pattern may also result from segregation or fragmentation of a large compact village. In this case, one or more sections of the village society choose or is forced to live a little away from the main cluster or village.
- In such cases, generally, the land-owning and dominant community occupies the central part of the main village, whereas people of lower strata of society and menial workers settle on the outer flanks of the village. **Such settlements are widespread in the Gujarat plain and some parts of Rajasthan. Hence statement 2 is correct.**

- **Dispersed Settlements**
- **The dispersed or isolated settlement pattern in India appears in the form of isolated huts or hamlets of few huts in remote jungles, or on small hills with farms or pasture on the slopes.**
- Extreme dispersion of settlement is often caused by the extremely fragmented nature of the terrain and land resource base of habitable areas. **Many areas of Meghalaya, Uttaranchal, Himachal Pradesh, and Kerala have this type of settlement. Hence statement 3 is correct.**

Q 85.B

- Maize (*Zea mays* L.) is the world's leading crop and is widely cultivated as cereal grain that was domesticated in Central America. **It is one of the most versatile emerging crops having wider adaptability. Globally, maize is known as the queen of cereals because of its highest genetic yield potential. Maize is the only food cereal crop that can be grown in diverse seasons, ecologies, and uses.**
 - Maize is a food as well as fodder crop grown under semi-arid climatic conditions and over inferior soils.
 - It is a Kharif crop that requires a temperature between 21°C to 27°C and grows well in old alluvial soil. **Hence statement 2 is correct.**
 - In some states like Bihar maize is grown in the rabi season also.
- **Maize cultivation is not concentrated in any specific region. Maize is cultivated in almost all types of agroecological regions.**
 - Only 7-8 states together account for more than three fourths of maize area as well as production in the country.
 - A regional shift in production has been observed from north to south; Bihar, Uttar Pradesh and Madhya Pradesh were the major maize producing states in 1990s, but during the past two decades, southern states, especially Andhra Pradesh and Karnataka, have become the major maize-producing states.
 - **Only four states: Karnataka, Rajasthan, Andhra Pradesh and Madhya Pradesh, constitute about half of the total maize acreage in the country, and five states, viz. Karnataka, Andhra Pradesh, Maharashtra, Rajasthan and Bihar, together account for about 2/3rd of the total maize production. Rajasthan and Madhya Pradesh though have a large share in maize area, but depict a low contribution to maize production in India. Hence statement 1 is not correct.**
- It is sown all over India and is now also introduced in Punjab and eastern and north-eastern regions.
- Maize is a food as well as fodder crop grown under semi-arid climatic conditions and over inferior soils.
- **Among the maize growing countries, India ranks 4th in area and 7th in production, representing around 4% of the world maize area and 2% of total production.**
- In India, maize is principally grown in two seasons, rainy (Kharif) and winter (rabi). **Kharif maize represents around 83% of maize area in India, while rabi maize corresponds to 17% of maize area.**

Productivity Group	Criteria	States	Percentage of Total Area Under Maize in India
High maize productivity states	> 4 T/Ha	Andhra Pradesh & Tamil Nadu	13%
Medium maize productivity states	2- 4 T/Ha	Bihar, Himachal Pradesh, Karnataka, Maharashtra, Odisha, Punjab, West Bengal.	40.2%
Low maize productivity states	< 2 T/Ha	Chhattisgarh, Gujarat, J&K, Jharkhand, MP, Rajasthan, UP.	43.8%
Other coarse cereals	16 Mn MT	1 Mn MT	0

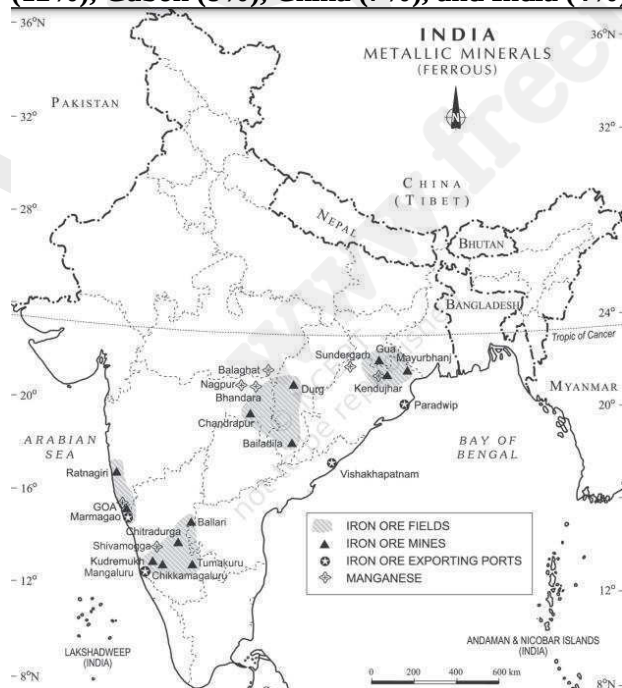


Q 86.A

- **Plantations** are a type of commercial farming where single crops of tea, coffee, sugarcane, cashew, rubber, banana or cotton are grown. A large **amount of labor and capital are required. Managerial and technical support, scientific methods of cultivation. The produce may be processed on the farm itself or in nearby factories.** The development of a transport network is thus essential for such farming. Major plantations are found in the tropical regions of the world. Rubber in Malaysia, coffee in Brazil, tea in India, and Sri Lanka are some examples.
- It was introduced by colonial powers across the regions. For instance, the French introduced cocoa and coffee plantations in west Africa. British introduced tea plantations in India and Sri Lanka, rubber plantations in Malaysia, and sugarcane and banana plantations in West Indies.
- **Hence option (a) is the correct answer.**

Q 87.A

- **Manganese is an important raw material for smelting iron ore and is also used for manufacturing ferroalloys.** It is also used for the manufacture of bleaching powder, insecticides, paints, glazed pottery, matches, batteries, and china clay. High amount of phosphorous makes the manganese ore unsuitable for its metallurgical use, whereas, high phosphorous and high iron contents make it unsuitable for Battery Industry. Manganese has no satisfactory substitute in its major applications. **Hence statement 1 is correct.**
- **Manganese deposits are found in almost all geological formations, however, it is mainly associated with the Dharwar system. Hence statement 2 is not correct.**
- The production of manganese ore at 2.90 million tonnes in 2019-20. Of the total production of manganese ore in 2019-20, **Madhya Pradesh continued to be the largest producer by contributing 33%, followed by Maharashtra (25%) and Odisha (19%).** The remaining production was reported by Andhra Pradesh, Karnataka Jharkhand, Rajasthan, Gujarat, and Telangana.
- **Statewise, Odisha tops the total reserves/ resources with 44% share followed by Karnataka at 22%, Madhya Pradesh 12%, Maharashtra & Goa at 7% each, Andhra Pradesh at 4%, and Jharkhand at 2%. Rajasthan, Gujarat, Telangana, and West Bengal together shared the remaining 2% of resources.**
- Odisha is one of the leading producers of Manganese. Major mines in Odisha are located in the central part of the iron ore belt of India, particularly in **Bonai, Kendujhar, Sundergarh, Gangpur, Koraput, Kalahandi, and Bolangir. Hence statement 3 is not correct.**
- Karnataka is another major producer and here the mines are located in Dharwar, Ballari, Belagavi, North Canara, Chikmagalur, Shivamogga, Chitradurga, and Tumakuru.
- Maharashtra is also an important producer of manganese, which is mined in Nagpur, Bhandara, and Ratnagiri districts. The disadvantage to these mines is that they are located far from steel plants.
- The manganese belt of Madhya Pradesh extends into a belt in Balaghat-Chhindwara-Nimar-Mandla and Jabua districts. Telangana, Goa, and Jharkhand are other minor producers of manganese.
- **In world, major manganese reserves are located in South Africa (32%), Brazil (17%), Australia (12%), Gabon (8%), China (7%), and India (4%).**

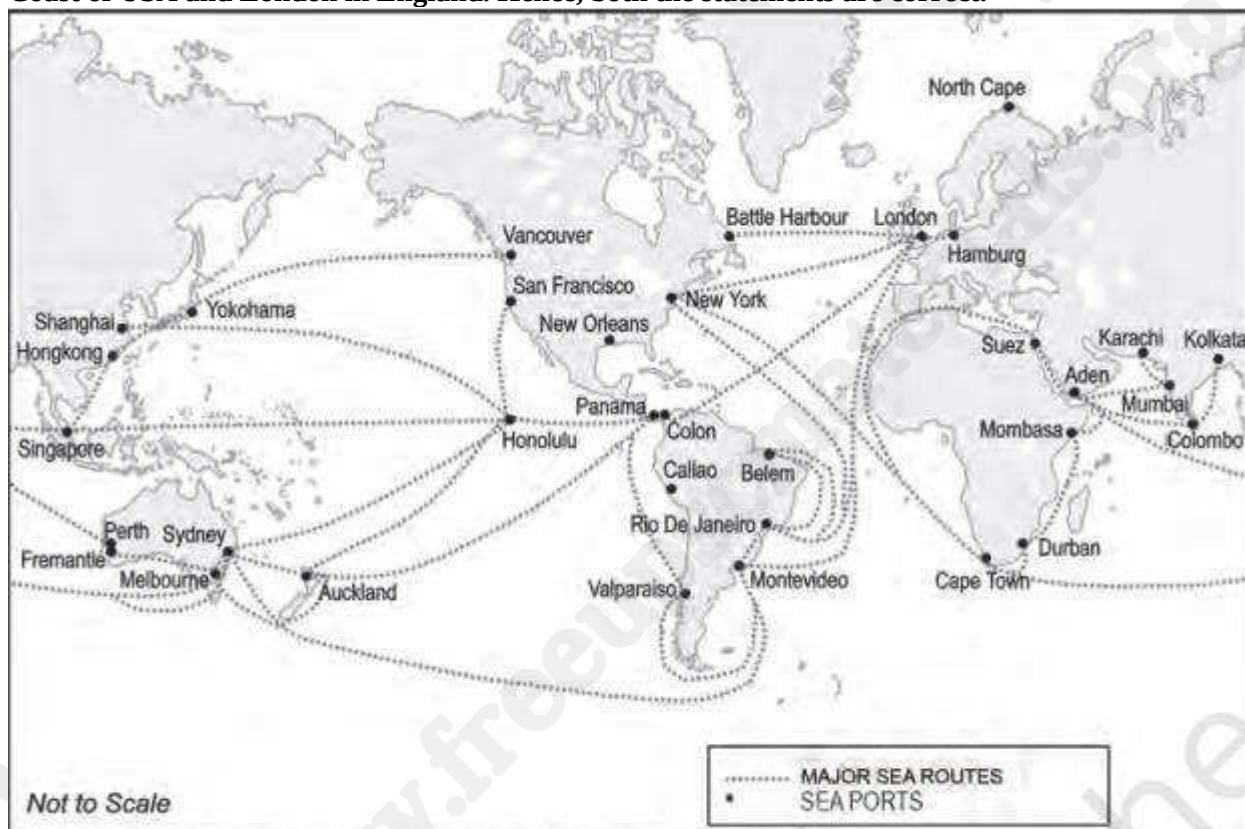


Q 88.D

- Primitive subsistence agriculture is also known as slash and burn agriculture. It is practiced by many tribes in the tropics. In Africa, south and central America, and South East Asia. It is known by different names across the regions. Jhumming in northeast India, Milpa in Central America, and Ladang in Indonesia and Malaysia. **Hence pair 1 is not correct.**
- Mixed farming is found in highly developed parts of the world. E.g., North-western Europe, Eastern North America, parts of Eurasia and temperate latitudes of Southern continents. Mixed farms are moderate in size and usually, the crops associated with them are wheat, barley, oats, rye, maize, fodder, and root crops. Crop rotation and intercropping play an important role in maintaining soil fertility. Equal emphasis is laid on crop cultivation and animal husbandry. Animals like cattle, sheep, pigs, and poultry provide the main income along with crops. **Hence pair 2 is not correctly matched.**
- Collective farming was introduced in the erstwhile USSR. Farming was based on social ownership. It was known as Kolkhoz. **Hence pair 3 is not correctly matched.**

Q 89.C

- **The Northern Atlantic Sea Route:** This links North-eastern U.S.A. and North-western Europe, the two industrially developed regions of the world. The foreign trade over this route is greater than that of the rest of the world combined. One fourth of the world's foreign trade moves on this route. It is, therefore, the busiest in the world and otherwise, called the **Big Trunk Route**. Both the coasts have highly advanced ports and harbour facilities. One of the most important sea link is between **New York in North Eastern Coast of USA and London in England. Hence, both the statements are correct.**



The International Boundary shown in this map may not to be considered as authentic

Fig. 8.9: Major Sea Routes and Sea Ports

Q 90.A

- **Recently, the Government emphasized promoting the sunflower area and its production in the country.**
- Sunflower is an important oilseed crop grown in temperate countries.
 - A major source of vegetable oil in the world.
 - India is one of the largest producers in the world.
 - Six states with **Karnataka in the lead** are the major producers of sunflower in the country. Karnataka with a production of 3.04 lakh tonnes from an area of 7.94 lakh hectares followed by Andhra Pradesh, Maharashtra, Bihar, Orissa, and Tamil Nadu are major sunflower producing states of India. **Hence statement 3 is not correct.**

- In India, cultivated over an area of 4.006 lakh ha with a production of 2.840 lakh tonnes and productivity of 709 kg/ha (Average of 2014-15 to 2018-19).
- Sunflower's contribution to Indian oilseed production is about 3.85%. **Hence statement 1 is correct.**
- The crop requires a cool climate during germination and seedling growth. It requires warm weather from the seedling stage up to the flowering stage and warm and sunny days during flowering to maturity.
 - High humidity accompanied by cloudy weather and rainfall at the time of flowering results in poor seed sets.
 - The amount of linoleic acid decreases with high temperatures at maturity.
 - **Sunflower is a photo-insensitive crop, therefore, it can be grown successfully in any season viz., Kharif, Rabi, and spring throughout India.** It takes about 80-90 days in Kharif, 105-130 days in Rabi, and 100-110 days in the spring season. **Hence statement 2 is correct.**
 - Sunflower, unlike most other crops, is not affected by the season and day length. With the exception of freezing temperatures, the sowing of sunflowers can be done in any month of the year
 - Sunflower can be grown on a wide range of soils and tolerates a moderate pH range and some salinity. It thrives best on deep loam soils with good drainage and irrigation facilities.
 - The optimum range of soil pH for this crop is 6.5 to 8.5.

Q 91.B

- **The new Industrial Policy was announced in 1991.** The major objectives of this policy were to build on the gains already made, correct the distortions or weaknesses that have crept in, maintain a sustained growth in productivity and gainful employment and attain international competitiveness.
- **Within this policy, measures initiated are:**
 - **abolition of industrial licensing,**
 - **free entry to foreign technology,**
 - **foreign investment policy,**
 - **access to the capital market,**
 - **open trade,**
 - **abolition of phased manufacturing program, and**
 - **liberalized industrial location program.**
- The policy has three main dimensions: liberalization, privatization, and globalization. The industrial licensing system has been abolished for all except six industries related to security, strategic or environmental concerns. At the same time, the **number of industries reserved for the public sector since 1956 has been reduced from 17 to 4.**
- **Industries related to atomic energy, substances specified in the Schedule of the Department of Atomic Energy as well as Railways have remained under the public sector.** The government also has decided to offer a part of the shareholdings in the public enterprises to financial institutions, the general public, and workers. **The threshold limits of assets have been scrapped and no industry requires prior approval for investing in the delicensed sector.** They only need to submit a memorandum in the prescribed format.
- **Hence option (b) is the correct answer.**

Q 92.D

- **Recent context:** Five of 12 public sector lenders are working with their technology service providers to go live on the AA system.
- **An Account Aggregator (AA) is a type of RBI regulated entity (with an NBFC-AA license)** that helps an individual securely and digitally access and share information from one financial institution they have an account with to any other regulated financial institution in the AA network. **Data cannot be shared without the consent of the individual.**
- **Aadhaar eKYC and CKYC only allow sharing of four 'identity' data fields for KYC purposes** (eg name, address, gender, etc). Similarly, **credit bureau data only shows loan history and/or a credit score.** The Account Aggregator network allows sharing of transaction data or bank statements from savings/deposit/current accounts.
- Account Aggregators cannot see the data; they merely take it from one financial institution to another based on an individual's direction and consent. **Contrary to the name, they cannot 'aggregate' your data. AAs are not like technology companies that aggregate your data and create detailed profiles of you.**
 - The data AAs share is encrypted by the sender and can be decrypted only by the recipient. The end-to-end encryption and use of technology like the 'digital signature' makes the process much more secure than sharing paper documents.

- **Registering with an AA is fully voluntary for consumers.**
 - If the bank the consumer is using has joined the network, a person can choose to register on an AA, choose which accounts they want to link, and share their data from one of their accounts for some specific purpose with a new lender or financial institution at the stage of giving 'consent' via one of the Account Aggregators.
 - A customer can reject the consent to share a request at any time.
- **Hence option (d) is the correct answer.**

Q 93.A

- **Recently, an indigenously developed helicopter launched Anti-Tank Guided Missile 'HELINA' was successfully flight-tested.**
- **About HELINA:**
 - **It is a Helicopter based NAG Missile. Hence statement 3 is not correct.**
 - **It is third generation fire and forget class anti-tank guided missile (ATGM) system mounted on the Advanced Light Helicopter (ALH). Hence statement 1 is correct.**
 - The system has all-weather day and night capability.
 - It can defeat battle tanks with conventional armour as well as explosive reactive armor.
 - Developed by: Defence Research and Development Organisation (DRDO).
 - It can engage targets in both indirect hit mode as well as top attack mode. In the top attack mode, the missile is required to climb sharply after launch and travel at a certain altitude then plunge to the top of the target. In the direct hit mode, the missile travels at a lower altitude directly striking the target. **Hence statement 2 is correct.**

Q 94.D

- **The basic indicator for employment is the proportion of the working-age population aged 15-59 who are employed (In some cases it may also be considered 15-64 years of age, therefore the important point to remember is that the working-age population begins from 15 years of age and not 18 years).**
- The working population (i.e. women and men of the age group – 15 to 59) take part in various occupations ranging from agriculture, forestry, fishing, manufacturing construction, commercial transport, services, communication, and other unclassified services.
- **The age dependency ratio is the ratio of dependents (people younger than 15 or older than 59) to the working-age population.** This indicator is measured as a percentage of the population.
- **Hence option (d) is the correct answer.**

Q 95.C

- Asia is the most populous continent in the world with 4.86 billion people accounting for nearly 59.5% of the world population. **But the population growth rate in Asia is 0.83%. Whereas Africa is the second largest continent by population with around 1.37 billion people. Africa has the highest population growth rate which is 2.45 %. Hence statements 1 and 2 are not correct.**
- Oceania with a population growth rate of 1.27% is the second-fastest-growing continent.

Q 96.B

- **Vedanthangal Bird Sanctuary:**
 - Located in **Tamil Nadu. Hence option (b) is the correct answer.**
 - The sanctuary was notified as RF (Reserve Forest) in 1963 under the Madras Act 1882.
 - Finally, in 1998, the sanctuary was notified under section 26(i) of the Wildlife Protection Act 1972.
 - There are about 28,000 land and water birds visiting this prestigious wetland sanctuary.
 - It is home to migratory birds such as pintail, garganey, grey wagtail, blue-winged teal, common sandpiper, and the like.
 - It is the oldest water bird sanctuary in the country.
 - Vedanthangal is the Tamil language that means 'hamlet of the hunter'.
- **Two new Ramsar sites:**
 - World Wetlands Day was celebrated on the 2nd of February 2022 across the globe. Two new Ramsar sites, Khijadia Wildlife Sanctuary in Gujarat and Bakhira Wildlife Sanctuary in UP were also announced on the occasion.
 - **Bakhira Wildlife Sanctuary in UP** provides a safe wintering and staging ground for a large number of species of the Central Asian Flyway while **Khijadia Wildlife Sanctuary (Gujarat)** is a coastal wetland with rich avifaunal diversity providing a safe habitat to endangered and vulnerable species.

Q 97.C

- **Permanent pasture** is the land used to grow grasses or other herbaceous forage naturally (self-seeded) or through cultivation (sown) and that is not included in the crop rotation of the holding for five years or longer. Most of this type of land is owned by the village Panchayat or the Government. Only a small proportion of this land is privately owned. **The land owned by the village panchayat comes under Common Property Resources. Hence statements 1 and 2 are correct.**
- **Current Fallow** : This is the land which is left without cultivation for one or less than one agricultural year. Fallowing is a cultural practice adopted for giving the land rest. The land recoups the lost fertility through natural processes.
- **Fallow other than Current Fallow** : This is also a cultivable land which is left uncultivated for more than a year but less than five years. If the land is left uncultivated for more than five years, it would be categorised as culturable wasteland.

Q 98.D



- **Rihand Dam** also known as Govind Ballabh Pant Sagar, is the largest dam in India by volume. The reservoir of Rihand Dam is called Govind Ballabh Pant Sagar and is India's largest artificial lake. Rihand Dam is a concrete gravity dam located at Pipri in Sonbhadra District in Uttar Pradesh, India. Its reservoir area is on the border of Madhya Pradesh and Uttar Pradesh. **It is located on the Rihand River, a tributary of the Son River.** The catchment area of this dam extends over Uttar Pradesh, Madhya Pradesh & Chhattisgarh whereas it supplies irrigation water in Bihar located downstream of the river.
- **The Hirakud Dam is located in the state of Orissa.** It is the longest dam in India with a total length of 25.79 km. It is also in the list of the longest dams in the world. The Hirakud Dam is situated on the river Mahanadi.
 - Height of the Dam- 61 mm, length of the Dam- 4.8 km (Main Dam), Type of Dam- Composite Dam, The reservoir capacity- is 47,79,965 acre, feet capacity that is installed- 347.5 Megawatt.
- **Mettur Dam** is a 1700 m long dam, across the **river Kaveri** was constructed in the year 1934. It provides irrigation facilities for 2.75 lakh acres of farmland in parts of Salem, the length of Erode, Namakkal, Karur, Tiruchirappalli, and Thanjavur districts.
- **Salal power station** is a run-of-the-river scheme with an installed capacity of 690 MW (Stage-I of 3 x115 MW & Stage-II 3 X 115 MW) to harness the Hydropower potential of river Chenab. It is located in the Reasi district of the Union Territory of Jammu & Kashmir.
- **Hence Option (d) is the correct answer.**

Q 99.B

- Nuclear energy has emerged as a viable source in recent times. Important minerals used for the generation of nuclear energy are uranium and thorium. **Uranium deposits occur in the Dharwar rocks.**
- Geographically, uranium ores are known to occur in several locations along the Singbhum Copper belt. It is also found in Udaipur, Alwar, and Jhunjhunu districts of Rajasthan, Durg district of Chhattisgarh, Bhandara district of Maharashtra, and **Kullu district of Himachal Pradesh. Hence statement 1 is not correct**
- The largest source of uranium comprises the monazite sands, both beach and alluvial. Some uranium is found in the copper mines of Udaipur in Rajasthan. **Hence statement 3 is correct.**
- **Thorium is mainly obtained from monazite and ilmenite in the beach sands along the coast of Kerala and Tamil Nadu.** The world's richest monazite deposits occur in the Palakkad and Kollam districts of Kerala, near Vishakhapatnam in Andhra Pradesh, and the Mahanadi river delta in Odisha. **Hence statement 2 is not correct.**

Table 2: Total Resources of Ilmenite (In Million Tonnes)

State	Total Insitu	State	Total Insitu
AP	163.05	Maharashtra	3.74
Jharkhand and Bihar	0.73	Odisha	96.44
Gujarat	2.77	Tamil Nadu	179.02
Kerala	145.70	West Bengal	2.05
		Total	593.50

State	Monazite (Million tonnes)
Odisha	2.41
Andhra Pradesh	3.72
Tamil Nadu	2.46
Kerala	1.90
West Bengal	1.22
Jharkhand	0.22
Total	11.93

- Australia has a substantial part (about 31 percent) of the world's uranium, Kazakhstan 12 percent, and Canada and Russia 9 percent each.
- **India has relatively modest reserves of uranium. Andhra Pradesh followed by Jharkhand and Meghalaya in that order is the top state with the largest uranium reserves.** With the rise in the number of reactors, India expects to import an increasing proportion of its uranium fuel needs. In 2013, India imported about 40% of its uranium requirements from France, Russia, and Kazakhstan.

Q 100.C

- Conurbation refers to a specific kind of geographical region. Due to the rapid increase in population and industrial and technological development, **the city boundary expands and one urban center coalesces with another** in a slow but continuous process of urbanization and regional development. The word conurbation has emerged from the 'continuous' and 'urban area'. **The word was used by Patrick Geddes in 1915 with reference to a continuous urban area of more than two urban centers which may have separate territorial units.**
- **Hence option (c) is the correct answer.**